

# Data Science in Finance
# DSF182

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# Module purpose and outcomes

**Module purpose**

This module is suitable for students who are interested in a career as a data scientist in the Finance industry. Students will be introduced to the type of data used in Finance, which is one of the areas in Commerce that deals specifically with large amounts of data. Students will be exposed to practical examples of day-to-day data science applications in finance, such as risk analytics and management, customer experience and management, consumer analytics and segmentation, personalised services, pricing automation, fraud detection, sentiment analysis, algorithmic trading, real-time analytics, predictive and forecasting analytics, etc. They will also develop an understanding of the general responsibilities of a data scientist in the finance sector. Lastly, they will learn to use appropriate techniques to identify key problems and propose data-driven solutions.

**Module outcomes**

Upon successful completion of this module, you will be able to:

1. Demonstrate knowledge of the structures, regulation and key systems/data vendors of the Finance industry.
2. Demonstrate an understanding of different Finance areas relevant to data science.
3. Demonstrate an understanding of general responsibilities of a data scientist in the Finance sector.
4. Evaluate challenges of working with big data in the field of Finance.
5. Use suitable techniques to identify key problems and select relevant approaches to finding solutions, given cases from the finance industry

The following topics will be covered in this study guide:

- Topic 1: *Structures, regulation and key systems / key data vendors of the finance industry*
- Topic 2: *Areas of finance relevant to data science – An overview*
- Topic 3: *General responsibilities of a data scientist in the Finance sector*
- Topic 4: *Big data technologies and challenges*
- Topic 5: *Techniques to identify key problems*
- Topic 6: *Practical cases of specific data scientist tasks in the Finance industry*

## Note

This Study Guide aims to help you navigate your prescribed textbook. It does not replace the textbook and therefore it cannot be used on its own to study the content of this module. In addition

to the prescribed textbook, recommended readings may be provided to enrich your learning. Although these are not compulsory, you are encouraged to consult them.

## Prescribed reading

The following prescribed textbooks are compulsory for this module:

Aldridge, I. and Avellaneda, M., 2021. Big data science in finance. John Wiley & Sons.

# Topic 1
# Structures, regulation and key systems / key data vendors of the finance industry

## 1.1 INTRODUCTION

This topic relates to the following module outcome(s):

1. Explain the structure, regulatory environment, and key information systems and data vendors of the finance industry.

To work meaningfully with financial data, a data scientist needs to understand the part of the finance industry from which the data originates. Finance contains several sub-sectors, each with different products, processes, regulatory duties and data. A model designed to assess a retail loan, for example, addresses a different problem from a model used to estimate an insurance claim or analyse a portfolio.

This topic provides an overview of the principal finance sub-sectors, the South African regulatory environment, and the systems through which financial data is recorded and exchanged. It also introduces fintech and blockchain, as these technologies influence how financial services and their supporting data systems are designed.

In this topic, you will gain knowledge in the following areas:

- The main sub-sectors of the finance industry
- The main business areas of banking, insurance, and asset management
- The rise of fintech and blockchain technology
- Key finance regulation, including FICA and FAIS
- Core enterprise systems (SAP, Oracle, SWIFT) and the problem of legacy systems
- Cloud storage and the role of data vendors such as Bloomberg and Thomson Reuters

## 1.2 SUB-SECTORS OF THE FINANCE INDUSTRY

For the purposes of this module, the finance industry is considered in terms of banking, insurance and asset management. These categories are not exhaustive, but they provide a

useful starting point. Banks accept deposits and provide credit and payment services; insurers pool and price risk; and asset managers invest funds on behalf of clients. Private equity, venture capital, payments and market infrastructure may be treated as separate areas in other classifications.

Fintech is best understood as a technology-driven layer that operates across these areas rather than as a single, separate type of institution. A payment application, digital lender or automated investment service may use different data and operate under different rules, even though each is described as fintech. Identifying the business area is therefore an important first step when deciding what a dataset represents and how it may be used.

**Read**

**Supporting:** Cao, L. (2023), "AI in Finance: Challenges, Techniques, and Opportunities", section 2.1, pp. 3 - 5 of the freely available preprint. Use the section to compare the principal financial businesses and the decisions they support.

## 1.3 MAIN BUSINESS AREAS OF BANKING

Banking may be divided into retail, commercial, corporate and investment-banking activities. Retail banking serves individuals through products such as transaction accounts, cards, personal loans and mortgages. Commercial banking serves small and medium-sized businesses, whilst corporate and investment banking provides services such as large-scale financing, securities issuance, advisory services and trading (Datoo, 2019, Chapter 1). The categories describe business areas and customer groups; they are not necessarily separate banks.

The kinds of data also differ across these banking areas. A retail bank records a continuing stream of payments, card authorisations and digital interactions. Corporate and investment banking usually involves fewer transactions, but an individual transaction may draw on contracts, financial statements and several market-data inputs. It is therefore the business problem, rather than the word 'banking', that determines how the data should be prepared and modelled.

**Read**

**Recommended:** Datoo, A. (2019), Legal Data for Banking, Chapter 1, pp. 1–10. Focus on how banking activities create records and why the legal and regulatory purpose of those records matters.

## 1.4 MAIN BUSINESS AREAS OF INSURANCE

Insurance pools the financial consequences of defined risks across a group of policyholders. Life insurance provides cover associated with events such as death or disability, whilst non-life insurance covers risks relating to property, vehicles, liability and other losses. Insurers assess applications, price cover, manage policies and evaluate claims (Corlosquet-Habart & Janssen, 2018).

These activities produce structured data, which follows predefined fields such as policy numbers, premium amounts and claim dates, as well as unstructured data whose useful meaning is not already stored in fixed fields, such as assessor reports, customer correspondence and photographs. Data science may support underwriting (assessing whether and on what terms to provide cover), claims triage (prioritising claims for review), fraud detection and loss forecasting. The volume and structure of the data depend on the type of insurance and the technology used; they should not be assumed to be uniform across the sector.

**Read**

**Recommended:** Corlosquet-Habart, M. and Janssen, J. (eds) (2018), Big Data for Insurance Companies, Chapter 1, pp. 1–25. Focus on insurance activities, the data they produce and the sector's uses of analytics.

## 1.5 MAIN BUSINESS AREAS OF ASSET MANAGERS

Asset managers invest funds on behalf of individuals and institutions according to an agreed objective, time horizon and level of risk. Their work includes investment research, portfolio construction, performance measurement and risk monitoring. Mutual funds, hedge funds and pension-fund portfolios are examples of structures within which these activities may occur.

Quantitative methods have been used in asset management for many years. Price histories, trading volumes, company information and economic indicators may be combined to evaluate securities and portfolios. Alternative data (such as satellite images of car parks, web traffic, news text or anonymised transaction trends) may add useful information, but its relevance and lawful use still need to be established (Aldridge & Avellaneda, 2021).

**Read**

**Supporting:** Ganti, A (2025). What Is Asset Management, and What Do Asset Managers Do?

## 1.6 THE RISE OF FINTECH AND BLOCKCHAIN TECHNOLOGY

Fintech refers to the use of technology to provide or support financial services. It includes digital payments, online lending, automated investment services and technology-enabled insurance, and it may be provided by a technology company, a partnership or a licensed financial institution. Phadke (2020) places this development within a broader progression in which data, connectivity, artificial intelligence and cloud technology increasingly shape financial products and operations.

Blockchain is one form of distributed-ledger technology. It allows authorised participants to share and update a record of transactions according to agreed rules. Cryptographic hashes make an alteration to recorded data detectable, whilst digital signatures allow a participant to prove that a transaction was authorised with the relevant private key. Financial applications include payment and settlement, tokenised assets, shared financial records and smart contracts. The recommended collection surveys examples involving cryptocurrency payments, security tokens, distributed financial data, insurance contracts and peer-to-peer lending (Dandapani & Shahrokhi, 2020). These applications create new data, but they also introduce questions relating to governance, identity, key management, data quality and responsibility when an automated rule is triggered.

**Read**

**Recommended:** Phadke, S. (2020), FinTech Future, Chapter 1, pp. 2–9, for the development of fintech; and Dandapani, K. and Shahrokhi, M. (eds) (2020), Blockchain Applications in Finance, guest editorial, pp. 709–713, for a map of blockchain applications in finance.

## 1.7 MAIN PIECES OF FINANCE REGULATION (FICA, FAIS, AND OTHERS)

Financial regulation influences what data may be collected, how it is processed and which records must be retained. In South Africa, the Financial Intelligence Centre Act 38 of 2001 (FIC Act) requires accountable institutions, meaning the businesses and professions specified in the Act, to implement a risk-based compliance programme. Relevant provisions cover customer due diligence, record keeping, suspicious-transaction reporting and the documented risk-management and compliance programme (Financial Intelligence Centre Act 38 of 2001, sections 20A–21H, 22–24, 29 and 42).

The Financial Advisory and Intermediary Services Act 37 of 2002 (FAIS) regulates specified financial advisory and intermediary services. Financial services providers, key individuals and representatives must meet the applicable requirements. Chapters 2–6 of the Determination of Fit and Proper Requirements for Financial Services Providers cover honesty and integrity, competence, continuing professional development, operational ability and financial soundness (Financial Sector Conduct Authority, 2017). Data-science tools may support an authorised professional by organising evidence, estimating risk or identifying cases for review, but responsibility for regulated advice or an intermediary service remains with the authorised provider and representative. POPIA is also relevant

whenever personal information is processed, and section 71 places conditions on certain decisions based solely on automated processing. These laws do not prescribe one data-science method, but they affect how systems are documented, monitored and reviewed.

Read
**Supporting:** Financial Intelligence Centre Act 38 of 2001, sections 20A–21H, 22–24, 29 and 42; Financial Advisory and Intermediary Services Act 37 of 2002, sections 7–8 and 13;), and POPIA, sections 8–19 and 71.

*It's not necessary to know these documents by heart, but for anyone working in the finance sector in South Africa, it's good to have a working knowledge of these acts.*

## 1.8 KEY SYSTEMS: SAP, ORACLE AND SWIFT

Regulation establishes the boundaries within which financial work takes place. The discussion now turns to the systems through which those regulated activities and their data are recorded and exchanged.

Large financial organisations use several types of operational system. SAP took its name from the German *Systemanalyse Programmentwicklung*, translated as "System Analysis Program Development", and is widely associated with enterprise resource planning (ERP): integrated applications that support processes such as accounting, procurement and human resources. Oracle is a company name rather than an acronym; it is known for database technology and also supplies cloud and enterprise applications. SAP and Oracle therefore overlap as enterprise-software providers, but "SAP" often refers to an integrated business application environment, whereas "Oracle" may refer either to an Oracle database or to one of its business applications. Core banking, policy-administration, trading and customer-management functions may still be provided by separate specialist systems.

The Society for Worldwide Interbank Financial Telecommunication (SWIFT) is a member-owned network through which financial institutions exchange secure, standardised financial messages. It carries payment instructions and other financial messages, but it does not itself hold customer accounts or transfer the underlying funds. Standard message formats can assist integration; however, they do not remove the need to validate meaning, completeness and access rights before the data are analysed.

**Read**

**Supporting:** SWIFT (n.d.), "What is SWIFT?". Read the description of its secure financial-messaging role and note that SWIFT does not itself hold funds or settle the underlying transaction.

## 1.9 THE PROBLEM OF LEGACY SYSTEMS IN BIG FINANCE ORGANISATIONS

A legacy system is an older system that remains important to current operations. Its age alone is not necessarily the problem. “Knowledge is scarce” may mean that the original developers have left, and field definitions or undocumented batch processes are understood by only a few employees. “Definitions differ” may mean that one system records a customer as a person, another records a household and a third records a separate customer entry for each product. Interfaces may be limited, and even a small change may disrupt payments, policy administration or another critical service. In large financial organisations, systems have often developed separately across products and business units, which may result in duplicated data and inconsistent identifiers (Datoo, 2019).

For a data scientist, much of the work may therefore take place before modelling starts. Fields must be traced to their source, customer or product identifiers reconciled, and old definitions compared with current ones. Replacing the legacy platform does not make this problem disappear; records, controls and service continuity must be preserved while the data and processes are moved.

**Read**

**Recommended:** Datoo, A. (2019), Legal Data for Banking, Chapter 4, pp. 53–80. Focus on data modelling, governance, business definitions and the consequences of fragmented systems.

## 1.10 CLOUD STORAGE

Legacy systems explain many of the constraints faced by financial institutions. Cloud services offer a different way of providing infrastructure, although they introduce dependencies and controls of their own.

Cloud computing allows computing, storage and software resources to be obtained as services and adjusted as demand changes. By contrast, on-premises infrastructure is hosted and managed within facilities controlled by the institution. Cloud services may support analytical storage, model development and scalable processing without requiring the institution to own every component of the underlying infrastructure.

Cloud use does not remove governance responsibilities. Financial institutions need to consider security, resilience, provider concentration (dependence on one or a small number of providers), access control, data location, offshoring (storing or processing data in another country), recovery and the ability to exit or change providers. South African financial regulators expect cloud and offshoring risks to be incorporated into governance, risk-

management and reporting processes (Financial Sector Conduct Authority & Prudential Authority, 2025).

| Deployment approach | Potential strengths for data science | Potential limitations |
|---|---|---|
| Cloud services | Rapid scaling; managed analytical services; flexible capacity; easier experimentation across approved environments. | Provider dependency; data-location and offshoring concerns; recurring cost; exit, connectivity and configuration risks. |
| On-premises infrastructure | Direct institutional control; may suit sensitive workloads or existing systems; predictable integration with some internal platforms. | Large up-front investment; slower scaling; internal maintenance burden; capacity may remain unused or become insufficient. |

**Read**

**Supporting:** Financial Sector Conduct Authority and Prudential Authority (2025), Joint Communication 2 of 2025: Cloud Computing and Data Offshoring, section 3.6 and sections 4.1–4.4, pp. 2–3. Focus on definitions, governance, risk appetite, confidentiality, resilience, legal requirements and due diligence.

## 1.11 DATA VENDORS: BLOOMBERG AND THOMSON REUTERS

Financial data scientists may combine internal operational data with information obtained from specialist market-data providers. Bloomberg, for example, supplies market data, news, analytics and communication tools through the Bloomberg Terminal and related services. Reuters historically supplied financial-market information, and its financial-data business became part of Refinitiv. Following LSEG's acquisition of Refinitiv in 2021, the Refinitiv brand was retired, and its services became part of LSEG Data & Analytics. Eikon was subsequently withdrawn and replaced by LSEG Workspace (LSEG, n.d.).

Vendor services can reduce the effort required to collect and standardise data from exchanges, company disclosures and other sources. They do not make the data automatically suitable for every analysis. Entitlements, meaning the permissions governing which data a user may access, and licensing terms still need to be understood. The same applies to identifiers, timestamps, missing observations and corporate actions such as dividend payments, share splits and mergers, which can affect how market data should be interpreted or adjusted. Internal data provides institution-specific context, whilst vendor data may provide the wider market context required for comparison.

### Activity

Choose one finance sub-sector, banking, insurance or asset management, and consider a data-science project such as predicting loan default, identifying potentially fraudulent insurance claims or evaluating investment risk. In approximately 300 words: (1) identify the internal operational data and external market data required; (2) explain which systems or services might provide them; (3) identify one regulatory or governance consideration; and (4) describe one data-quality or integration problem.

***Indicative answer:*** A suitable response will connect a clearly defined business decision to relevant internal and external data; distinguish operational systems from external providers; identify an applicable requirement such as FIC Act monitoring, FAIS responsibility or POPIA; and explain a realistic problem such as inconsistent identifiers, changed definitions, access restrictions or missing vendor observations.

## Recommended Reading

The following optional readings are not compulsory but are recommended to enrich your understanding of this topic. Full publication details and access links appear in the consolidated References section.


Dandapani, K. and Shahrokhi, M. (eds) 2020. Blockchain Applications in Finance. Bingley: Emerald. https://www.proquest.com/legacydocview/EBC/6423594?accountid=198890.

Datoo, A. 2019. Legal Data for Banking: Business Optimisation and Regulatory Compliance. Hoboken, NJ: Wiley. https://www.proquest.com/legacydocview/EBC/5761270?accountid=198890.

Phadke, S. 2020. FinTech Future: The Digital DNA of Finance. New Delhi: SAGE. https://www.proquest.com/legacydocview/EBC/6121183?accountid=198890.

Smith, D. and El-Wakeel, F. 2019. "What is a data science model?", Strategic Finance, 101(5), pp. 60–61. https://www.proquest.com/scholarly-journals/what-is-data-science-model/docview/2313318329/se-2?accountid=198890


## Summary

This topic introduced the business and technical context within which financial data is produced. Banking, insurance and asset management perform different functions and consequently generate different combinations of transactional, customer, claims and market data. Fintech operates across these areas and uses technology to change how services are delivered. Blockchain is one example of such technology, with possible applications in shared records, tokenised assets and rules-based contracts. These applications should be evaluated in terms of both their usefulness and the governance questions they introduce.

The regulatory environment affects the collection and use of financial data. In South Africa, the FIC Act establishes risk-based duties relating to customer due diligence, records, monitoring and reporting. FAIS regulates specified advisory and intermediary services and imposes applicable fit-and-proper competence requirements on providers, key individuals and representatives; analytical tools may support these professionals but do not assume responsibility for regulated advice. POPIA governs personal information and places conditions on certain solely automated decisions. These requirements influence system and model design, but they do not prescribe a single technical solution.

Financial data is distributed across operational, specialist and legacy systems. SAP and Oracle provide overlapping enterprise technologies, whilst core banking, policy, trading and customer functions may use other platforms. SWIFT provides secure financial messaging rather than moving the underlying funds. Cloud services can provide scalable storage and processing, but institutions remain responsible for security, resilience, offshoring and provider concentration. External vendors such as Bloomberg and LSEG provide market information that may be combined with internal data. In every case, a data scientist still needs to establish what the data represents, how it was produced and whether it is suitable for the intended analysis.

## Self-Assessment Questions

The following questions are not graded. Use them to test your understanding of the key concepts covered in this topic before moving on. Try to answer each question fully in your own words before checking the suggested answers at the back of this study guide.

1. Identify the main sub-sectors of the finance industry and briefly describe the kind of data each is likely to generate.
2. Explain why understanding which finance sub-sector a dataset comes from matters to a data scientist, using an example from banking and an example from insurance.
3. Discuss two ways in which blockchain technology is being applied in finance, and explain what kind of data science opportunity each application creates.
4. Explain the purpose of FICA and FAIS, and discuss how these pieces of regulation constrain the way a data scientist may work with client data.
5. Describe the problem of legacy systems in large finance organisations, and explain why this problem is particularly significant for data scientists.
6. Compare the role of an internal enterprise system such as SAP with the role of an external data vendor such as Bloomberg in supplying data for a finance data science project.

# Topic 2
# Areas of finance relevant to data science – An overview

## 2.1 INTRODUCTION

This topic relates to the following module outcome(s):

1. Demonstrate insight into the range of areas within finance in which data science is applied.

Having established the structure of the finance industry in Topic 1, this topic considers where data science may support financial work. The focus is on the business problem addressed in each area rather than on the mathematical detail of a particular algorithm. The prescribed textbook provides technical examples, particularly in forecasting, trading and asset pricing, whilst the additional readings extend the discussion to insurance and operational applications.

The areas introduced below frequently overlap. A fraud system may operate in real time, use predictive modelling and influence a customer's experience during the same transaction. Separating the areas remains useful for study, but a data scientist should also be able to explain how they work together within a financial process.

In this topic, you will gain knowledge in the following areas:

- Risk analytics and management
- Customer experience, consumer analytics and personalisation
- Pricing automation and fraud detection
- Sentiment analysis and algorithmic trading
- Real-time analytics and predictive and forecasting analytics

## 2.2 RISK ANALYTICS AND MANAGEMENT

Risk analytics uses data to identify, measure and monitor uncertainty that may affect a financial institution or its clients. A borrower is a person or organisation that receives funds and must repay them. A counterparty is the other party to a financial contract or transaction, such as a bank on the other side of a derivatives trade. The principal risk categories below use different evidence and support different decisions.

| Risk type | Meaning | Illustrative data and example |
|---|---|---|
| Credit risk | A borrower or counterparty may not meet an obligation. | Repayment and affordability data; estimating whether a home-loan applicant is likely to default. |
| Market risk | Losses may arise from changes in prices or other market factors. | Positions, prices and volatility; estimating the effect of an interest-rate change on a bond portfolio. |
| Operational risk | Losses may arise from failed processes, people or systems, or from external events. | Incident, control and scenario data; identifying repeated payment-processing failures. |
| Liquidity risk | An institution may be unable to obtain sufficient cash or readily available funds when obligations fall due. | Cash-flow and funding data; forecasting whether daily payment obligations can be met. |

**Read**

**Optional:** Fernando, J. (2026), Understanding Derivatives: A Comprehensive Guide to Their Uses and Benefits. This is an optional bit of reading, but a good one if you are interested in learning more about derivatives.

## 2.3 CUSTOMER EXPERIENCE AND MANAGEMENT

Risk analytics is one major application area, but financial decisions also affect how customers experience and use services. The next three sections move from the overall service journey to group-level analysis and then to individualised decisions.

A customer may start an application on a mobile phone, contact a call centre when it fails and later visit a branch. Customer-experience analytics links these parts of the service journey so that repeated contacts, failed steps or signs that a customer may leave can be identified. For example, a bank could combine mobile-app error events with call-centre records, find where applicants repeatedly abandon digital account opening, and then test whether a redesigned step changes that outcome.

Customer experience concerns the quality and continuity of an individual service journey. The related concepts of consumer analytics, segmentation and personalisation are introduced in the following sections, where their different purposes and units of analysis are compared.

**Read**

**Optional:** Peddamukkula, P. K. (2023). The role of AI in personalization and customer experience in the financial and insurance industries.

## 2.4 CONSUMER ANALYTICS AND SEGMENTATION

Consumer analytics uses product, transaction, channel and demographic information to understand customer behaviour. Segmentation and personalisation differ in both their unit of analysis and their intended use. Segmentation may be seen as the process of splitting a larger population group into smaller groups in which the smaller population group share certain traits (such as an income range).

| Aspect | Consumer analytics | Segmentation | Personalisation |
|---|---|---|---|
| Main purpose | Understand behaviour and patterns | Identify useful customer groups | Adapt a service, message or recommendation |
| Unit of analysis | Population, market or customer base | Group of customers | Individual customer |
| Example | Examine digital-saving behaviour | Identify salary customers who seldom use savings tools | Select a message using one customer's behaviour |
| Main caution | Patterns still require interpretation | Do not assume that every group member is identical | Limit intrusive data use and test customer impact |

Clustering is one method that may be used for segmentation because it identifies groups without requiring a predefined class label (Aldridge & Avellaneda, 2021). The resulting groups still require interpretation. A mathematically distinct cluster is not automatically a useful or fair customer segment, and the institution should test whether it supports the intended decision.

**Read**

**Prescribed:** Aldridge, I. and Avellaneda, M. (2021), Chapter 9, pp. 262–269. Focus on the purpose of clustering and the conceptual distinction between K-means and spectral clustering. The detailed linear algebra is not required.

## 2.5 PERSONALISED SERVICES AND PRICING AUTOMATION

Personalisation adapts a service, message or recommendation using information about an individual customer. Pricing models may similarly estimate a premium, fee or interest rate from information associated with risk and expected cost. Insurance examples include using claims history or telematics data, such as information collected by a vehicle or connected device about its use, to inform the price of cover (Corlosquet-Habart & Janssen, 2018).

Consistency and speed are not the only considerations when these models are used. Historical data may contain earlier patterns of disadvantage, and a proxy variable may indirectly reveal a sensitive characteristic even when that characteristic was deliberately excluded. POPIA section 71 is particularly relevant where a decision is based solely on automated processing and has legal or substantial consequences. In such a case, testing and monitoring have to be supported by an understandable explanation and a route for human intervention.

## Multiple-choice questions

Select the best answer for each question. The correct answers appear after all four questions.

**1. Which example is most clearly a personalised financial service?**

A. Sending every customer the same interest-rate notice
B. Recommending a savings product using the customer's stated goal and circumstances
C. Publishing the bank's annual report
D. Applying one fixed fee to every account

**2. Why might vehicle-telematics data create an unfair pricing outcome?**

A. Telematics data can never be collected accurately
B. Location, travel time or mileage may act as proxies for circumstances outside the customer's control
C. Insurance prices may not use any customer data
D. Telematics always results in lower premiums

**3. Which safeguard is most appropriate when an automated decision may materially affect a customer?**

A. Remove all records of how the decision was produced
B. Allow the model to make the decision without review
C. Provide an explanation and a route for human review or correction
D. Use as many variables as possible

**4. What is the main purpose of customer segmentation?**

A. To place customers into useful groups based on relevant shared characteristics
B. To assume that every customer within a group is identical
C. To replace all individual customer records
D. To guarantee that a marketing campaign will succeed

**Correct answers:** 1—B; 2—B; 3—C; 4—A.

**Read**

**Recommended:** Corlosquet-Habart, M. and Janssen, J. (eds) (2018), Chapter 5, pp. 131–161, for data-driven insurance uses including pricing and claims. Also read POPIA section 71 before considering a solely automated decision about a person.

## 2.6 PREDICTIVE AND FORECASTING ANALYTICS

The discussion so far has described decisions made about customers. Many of those decisions rely on an estimate, but an estimate of an outcome and a forecast over time are not quite the same task.

Predictive analytics uses available information to estimate an unknown or future outcome. Forecasting is more specifically concerned with values over time, such as cash flow, losses, demand or market variables. These techniques establish useful foundations for later sections: fraud detection may predict whether a current transaction is suspicious; pricing may estimate an expected claim cost; and trading may forecast a market quantity over a defined horizon.

Not every application is a forecasting problem. Fraud detection may classify a transaction taking place now, customer segmentation identifies groups, and some pricing models estimate expected cost. The shared feature is that data is used to produce an estimate or structured decision that can be evaluated against a defined business objective.

**Read**

**Supporting:** Hyndman, R.J. and Athanasopoulos, G. (2021), Forecasting: Principles and Practice, sections 1.1, 1.3, 1.4 and 1.6. Focus on what can be forecast, the intended horizon and use, the choice of data and method, and the five basic steps in a forecasting task.

## 2.7 FRAUD DETECTION

Fraud detection may combine rule-based checks (for example, flagging an impossible change in location), statistical models such as logistic regression and machine-learning techniques such as random forests or XGBoost. These methods identify behaviour that differs from an expected pattern or resembles known fraud. Payment amount, location, device, timing and previous account activity may contribute to a score. Anti-money-laundering controls have a related but broader purpose: they seek to prevent criminals from disguising the origins of unlawfully obtained funds. In this context, accountable institutions are required to monitor transactions in accordance with a risk-based compliance programme (Financial Intelligence Centre, 2025).

False positives and false negatives create different problems. When a legitimate transaction is flagged, the customer may be inconvenienced and the bank incurs an investigation cost; when fraud is missed, the customer and institution may suffer a loss. Lowering a threshold generally finds more suspicious cases, but it can also block more genuine transactions. The choice of threshold is therefore a business and control decision. The model ranks transactions or cases for a response; it does not, on its own, prove that fraud occurred.

**Read**

**Supporting:** Cao, L. (2023), section 2.2, pp. 5–6, and Table 1, pp. 8–10, for anomaly and fraud-detection applications; and the Financial Intelligence Centre Act, sections 29 and 42, for suspicious-transaction reporting and a risk-based compliance programme.

## 2.8 SENTIMENT ANALYSIS

Natural language process is the use of a model to analyse plain text for specific factual elements, translation or sentiment analysis. Sentiment analysis uses natural language processing (NLP) to estimate whether text expresses a positive, negative or neutral position. In finance, the text may include news, company announcements, analyst commentary or social-media posts. The resulting measure may be used with price, volume or other information rather than treated as a complete trading signal on its own (Aldridge & Avellaneda, 2021).

**Read**

**Prescribed:** Aldridge, I. and Avellaneda, M. (2021), Chapter 4, pp. 80–85. Focus on financial news, sentiment, unstructured information and the analyst workflow. The later mathematical treatment is not required.

## 2.9 ALGORITHMIC TRADING

Algorithmic trading uses programmed rules to generate or execute orders on a stock exchange or another trading venue. A simple rule-based model might buy or sell when a price crosses a predefined threshold. A more complex forecasting model might estimate a short-term return or price movement and generate a trade when the expected opportunity exceeds transaction costs. Execution algorithms can then divide or schedule the order to reduce its market impact.

Both simple and complex approaches require testing, limits and monitoring. A rule may behave unexpectedly when market conditions change, and a forecasting model may learn a relationship that does not persist. The objective is not to automate trading at any cost, but to show how a defined market signal is converted into a controlled order and evaluated after fees, slippage and risk.

**Read**

**Optional:** Seth, S. (2026). Basics of Algorithmic Trading: Concepts and Examples.

*This is an interesting source if you want to learn a bit more about algorithmic trading.*

## 2.10 REAL-TIME ANALYTICS

Fraud monitoring and algorithmic trading both show that the timing of an output can be as important as the method used to produce it. This leads to the wider question of real-time analytics.

Real-time analytics processes incoming data and produces an output quickly enough to influence the event being observed. A card transaction may need to be assessed before authorisation, whilst a trading decision may depend on a much shorter interval. The meaning of 'real time' therefore depends on the business requirement.

Real-time systems require several connected stages to operate reliably. Data ingestion brings incoming data into the system; processing prepares it; model scoring applies a trained model to produce an estimate or classification; and the response stage uses that output. These systems must also address delayed or duplicated events, system failure and monitoring. A rapid result is only useful when the underlying data and decision process are sufficiently dependable.

**Read**

**Optional:** Aquino, J. (2025). What is real-time analytics?.

*There are many aspects related to real-time analytics. This source delves into a few of the basic considerations.*

# Recommended Reading

The following optional readings are not compulsory but are recommended to enrich your understanding of this topic. Full publication details and access links appear in the consolidated References section.

Corlosquet-Habart, M. and Janssen, J. (eds) 2018. Big Data for Insurance Companies. Hoboken, NJ: Wiley. https://www.proquest.com/legacydocview/EBC/5261379?accountid=198890.

Pokharkar, A.P. 2021. “Data science applications: Applications of data science with concrete real-life use cases”, ThinkQuest International Journal of Technology and Management, 6, pp. 42–47. https://mmbgims.com/wp-content/uploads/2021/04/TQIJTM_Issue-6_v3.pdf#page=42

# Summary

The application areas in this topic overlap more than their separate headings suggest. Risk, customer-experience, pricing and fraud decisions may all draw on prediction, while forecasting is concerned more specifically with values over time. Sentiment analysis adds evidence from text, algorithmic trading turns a defined signal into an order, and real-time analytics determines whether an output arrives soon enough to be useful. In each case, the method must follow the financial problem, the timing of the decision and the consequences of error.

# Self-Assessment Questions

The following questions are not graded. Use them to test your understanding of the key concepts covered in this topic before moving on. Try to answer each question fully in your own words before checking the suggested answers at the back of this study guide.

1. Distinguish between credit risk, market risk and operational risk, and give an example of a data source relevant to each.
2. Explain the difference between consumer segmentation and personalisation and describe how a bank might use each.
3. Discuss the key trade-off that must be balanced when designing a fraud detection system and explain why this trade-off matters to customers as well as to the institution.
4. Explain how sentiment analysis contributes a different type of information to a trading decision than traditional price and volume data.
5. Why is real-time analytics described as a technical requirement rather than an application area in its own right? Support your answer with an example.
6. Explain why predictive analytics and forecasting are related but not interchangeable. Use two application areas from this topic to support your answer.

# Topic 3
# General responsibilities of a data scientist in the Finance sector

## 3.1 INTRODUCTION

This topic relates to the following module outcome(s):

- Demonstrate an understanding of general responsibilities of a data scientist in the Finance sector.

Selecting an algorithm and reporting its accuracy are only part of data-science work. In finance, the data and software eventually support a business decision that may affect customers and is subject to regulation and internal control. A model can therefore look successful in an experiment and still fail in practice because the wrong problem was framed, the data cannot be traced or the output does not fit the operating process.

This topic considers the responsibilities that arise as work moves from data collection and analysis to prototyping, deployment and communication. Particular attention is given to mixed data sources, collaboration, validation and the need to protect personal information. These matters form part of technical quality because a result that cannot be explained, reproduced or used responsibly has limited value.

A large institution normally distributes these responsibilities across several roles. The data scientist may not operate every pipeline or approve every control, but still needs to understand where the work depends on other teams and where an assumption could be lost. Figure 3.1 shows this as a connected lifecycle: the financial decision and data foundation come first, followed by analysis, prototyping, validation and deployment. Monitoring and communication continue after release and may send the team back to an earlier stage.1w

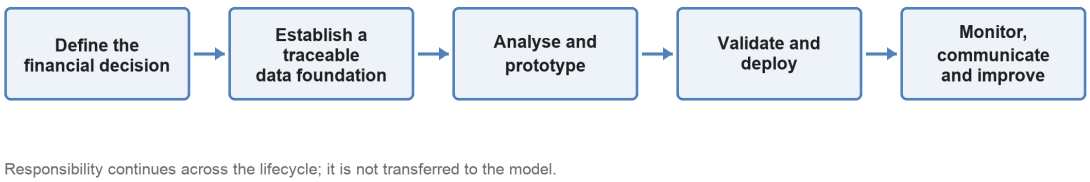



**Figure 3.1: Responsibilities across the financial data-science lifecycle**

*Source: Author*

In this topic, you will gain knowledge in the following areas:

- Collecting strategic data and designing documented data foundations
- Bringing cohesion to structured, semi-structured and unstructured data
- Using natural language processing (NLP) and computer vision in defined financial processes
- Working with technical, business, risk and management teams
- Turning quantitative analysis and prototypes into measurable solutions
- Managing models, analytical code, communication and professional responsibility

## 3.2 COLLECTING STRATEGIC DATA AND BUILDING THE DATA FOUNDATION

A request for *'customer data'* or *'transaction data'* sounds specific but usually is not. An application, an approved account, an attempted payment, a posted transaction and a complaint each describe a different part of a financial process. Before modelling starts, the data scientist has to establish which decision will use the output, what one observation represents, and which date or event fixes the point of analysis.

Although data acquisition is sometimes presented as a technical extraction task, available data are not necessarily suitable data. Before collection, it is necessary to establish who owns the data, why they were obtained, whether the proposed use is lawful, how frequently they are updated and which access controls apply. Unnecessary fields increase cost, privacy exposure and the number of possible quality problems.

Reproducibility involves more than running the same notebook for a second time. The pipeline has to show where the data came from, which transformations were applied and what happened when a source was late or incomplete. The relevant snapshot or extract date, code, parameter settings, exclusions and source-system versions also need to be retained. Together with field definitions, ownership and lineage, this record allows another practitioner to rebuild the analytical dataset and explain why two runs differ—a requirement that supports investigation and regulatory reporting as well as technical review.

### Did you know?

A field can keep the same name while its semantic meaning changes. For example, transaction_date might record the authorisation timestamp in one system but the settlement date after a migration. Analyses that compare periods across the change could then create false timing patterns unless the applicable definition and transformation are recorded.

**Read**

**Supporting:** Chang, W.L. and Grady, N. (2019), NIST Big Data Interoperability Framework: Volume 1, sections 5.1–5.3, pp. 25–30, and section 7.2, p. 36. Focus on the data-science process, data and metadata, and governance.

## 3.3 BRINGING COHESION TO STRUCTURED AND UNSTRUCTURED DATA

Financial data may be structured, semi-structured or unstructured. The distinction concerns how explicitly the data are organised before analysis.

| Data type | Typical structure | Financial example | Key consideration |
|---|---|---|---|
| Structured | Fixed rows, columns and data types | Account or transaction table | Definitions and keys must remain consistent |
| Semi-structured | Labels or markers without one fixed table | JSON payment message or application log | Fields may differ between records |
| Unstructured | Meaning is not stored as predefined fields | Complaint text, contract, image or audio | Interpretation may require NLP or computer vision |

Bringing these sources together requires a consistent representation without removing their context. Customer identifiers, dates and currencies may need to be reconciled, whilst a document must remain linked to the transaction or case to which it relates. Here, data modelling concerns the definition of entities, relationships, fields and rules; it does not refer only to fitting a statistical model.

A neat schema can still describe the wrong thing. Two records may refer to different events, and a missing value is not automatically a zero. Matching rules, aggregations and derived variables need to be checked against source records and discussed with people who know the business process. Any remaining uncertainty should remain visible in the documentation instead of being hidden by an apparently complete table.

**Read**

**Optional:** Gibson, R. 2026. Structured, Unstructured, and Semi-Structured Data: A Guide for the AI Era.

*Review this source to learn a bit more about the distinction between structured, unstructured and semi-structured data.*

## 3.4 USING NLP AND COMPUTER VISION

Natural language processing (NLP) and computer vision are specialisations within data science that work with language and visual information. NLP enables computers to represent, extract or classify meaning from written or spoken language. Computer vision enables computers to derive

information from images or video by detecting, locating or classifying visual features, objects and text. Financial applications include classifying complaints, reviewing contracts, extracting fields from identity documents and examining photographs submitted with insurance claims.

Language and image data need different preparation from ordinary tables. NLP may involve language handling, tokenisation and document-layout detection; computer vision may require image enhancement, orientation correction or object localisation. Evaluation also changes with the task. A document-extraction system may be assessed at word or field level, while an image system may require pixel-, region- or object-level measures. A single accuracy figure can hide failures affecting uncommon documents, languages, lighting conditions or customer groups.

A responsible workflow keeps the original material and records how labels were produced. In an insurance example, the test set would need more than clear claim photographs; blurred images, unfamiliar vehicle models and unusual lighting reveal whether the system works under realistic conditions. Where an error could deny a service or start an investigation, an uncertain result belongs with a human reviewer. The model output remains one item of evidence within the claims process.

**Read**

**Supporting:** James, G. et al. (2023), Chapter 10, section 10.3, pp. 406–412, and section 10.4, pp. 413–415, for conceptual introductions to convolutional neural networks and document classification. Detailed derivations are not required.

## 3.5 WORKING ACROSS BUSINESS AND TECHNICAL TEAMS

The NLP and computer-vision examples show that choices about labels, error and human review cannot be made from code alone. They depend on colleagues who understand the process, controls and affected users.

Data-science projects cross organisational boundaries because no single role holds all the required business, technical and governance knowledge.

| Role or team | Typical contribution |
|---|---|
| Business and operational specialists | Define the process, decision and practical constraints. |
| Data scientists | Frame the analytical problem and evaluate the evidence. |
| Data engineers and database administrators | Build and maintain data pipelines and stores. |
| DevOps and MLOps practitioners | Support controlled software and model release, monitoring and retraining. |
| Information security | Protect systems, credentials and sensitive data. |
| Legal, compliance, risk and validation | Interpret obligations and independently challenge assumptions and performance. |
| Management | Decide whether the expected benefit justifies the remaining risk. |

The data scientist often connects these perspectives by turning a broad request into a testable problem. “Improve fraud detection”, for example, could become a requirement to reduce confirmed card-fraud losses without allowing legitimate declines to exceed an agreed level. The reformulation makes both the expected benefit and the customer impact measurable.

Consider a field that is created only after a lending or fraud decision. A business specialist can regard it as highly relevant, while the data scientist includes it as a model input and obtains unusually good results. This is information leakage because the field did not exist at the time of the live decision. Evaluation data that are too similar to the training data create a related problem. Recording the timing of each candidate field, and checking it with operational colleagues, exposes both problems before deployment.

## 3.6 FROM QUANTITATIVE ANALYSIS TO A WORKABLE SOLUTION

Quantitative analysis turns a business question into evidence, but the evidence comes in different forms. Counts, loss summaries and customer profiles describe what happened; diagnostic work looks at why a pattern occurred; and predictive or experimental methods look at what may happen or what changes after an intervention. The method need not be complicated to be useful. An elaborate model adds little where a transparent rule or process correction answers the question reliably.

The analytical result is workable when it fits the surrounding process. The team must determine who receives the output, what happens next, how fast it is needed, and what happens if the system is unavailable. The fraud score could lead to an approval, a rejection or a manual review, each of which comes with different costs and consequences.

A good example for why technical and practical measures should be read together is the F1 score. Precision is the fraction of flagged cases that are relevant, recall is the fraction of relevant cases that were found, and F1 is a summary of the trade-off between the two. None of these measures account for losses avoided, review effort, complaints or customer disruption. To this end, performance still needs to be compared to a baseline and tracked as the population and operating process change.

### Did you know?

A fraud model can improve recall by flagging more transactions, but this may also increase false alerts and the workload of investigators. The technically “better” threshold therefore depends on loss, customer disruption and review capacity and not on one metric alone.

**Read**

**Supporting:** James, G. et al. (2023), Chapter 2, sections 2.1–2.2, pp. 15–39. Focus on supervised and unsupervised learning, regression and classification, interpretability and model accuracy. Use these distinctions to connect the business question to an analytical method and evaluation plan.

## 3.7 TRAINING MODELS AND PROTOTYPING SYSTEMS

An analytical design that works still needs to be tested. The next opportunity to challenge assumptions is model training and prototyping before committing to production.

The model training is trying to estimate patterns for new cases from existing examples. That sounds simple enough, but it needs to have a defensible target, relevant variables and a real difference between development and evaluation data. Old records may show an obsolete process, earlier discrimination or an odd event. Leakage can make performance then look much better than it is.

A prototype tests assumptions before the institution commits to a production system. It may indicate if data can be obtained, if performance is sufficient and output can be integrated into process. Prototypes include computational notebooks, which combine code, results and explanation It is useful for exploration, but does not inherently provide the security, monitoring, support, ownership and controlled documentation required in production.

Models need to be tested before use and monitored after use. Validation includes concept, data, implementation, performance and limitations, with the depth of review depending on the possible effect. Within supervised financial institutions, supervisory guidance may inform the validation standards of the institution thru an emphasis on documentation, governance, independent challenge and ongoing monitoring. It does not replace but supplements the institution's own model validation policy and approval process.

Read

**Optional:** Aravindan, A. 2024. Prototyping Stage in Financial Models Development.

*This source delves a bit deeper into the prototyping stage in the development of financial models.*

## 3.8 DEVELOPING ALGORITHMS AND MANAGING ANALYTICAL CODE

New code is sometimes taken as proof of innovation. In a financial application however, a new algorithm is only allowed into the fold when it solves a need that an established method cannot. Well-tested methods are usually easier to explain, validate and maintain. With a new method, the assumptions, limitations and added benefit need to be demonstrated.

Familiarity is not a defence either: re-using an unsuitable method can be just as difficult to justify as adding unnecessary complexity.

Analytical code is part of the evidence supporting a financial decision. Another practitioner should be able to see how a result was produced and what has changed from version to version. Version control (usually Git) keeps a history of changes to the code, and branches and pull requests are used to review work before merging. Testing confirms that the implementation works as expected. Reproduction also requires recording of package versions, data snapshots, configuration settings, and any random seed used in a stochastic procedure. Passwords, access credentials, and personal information should not be encrypted or in uncontrolled logs.

This is especially true as analytical work transitions from experimental to production. While an experimental environment may provide the data scientist with a great deal of freedom to explore alternative approaches, it does not provide the authority to release a model into an operational process.

Typically, getting a model into production requires independent review, formal approval and a controlled release. Segregation of duties is the process of making sure that one person cannot independently develop, approve and release a material model without a review process in place. The controls are not designed to stop experimentation. They are designed to make the final analytical process traceable, reproducible and accountable before it shapes a financial decision.

**Read**

**Supporting:** Sandve, G.K. et al. (2013), "Ten Simple Rules for Reproducible Computational Research", Rules 1 and 3–7. Focus on tracing how results were produced, recording dependencies, version-controlling scripts, retaining intermediate evidence and recording random seeds.

*Although this source is published in a biology journal, the general principles presented apply to finance systems too.*

## 3.9 COMMUNICATION, DOCUMENTATION AND PROFESSIONAL RESPONSIBILITY

Producing traceable code is not enough; that is the minimum of responsibility. The evidence must also be accessed by those who endorse, operate or review or are affected by the resulting process.

Different audiences have to act on the same evidence, and that makes communication part of the technical work. Executives will need the expected benefit and main risks, operational users the action and escalation route, reviewers enough detail to reproduce the work, and an affected customer may need an explanation that can be understood without knowing how the model was built.

Documentation should identify the purpose and scope of the work, the data and assumptions used, the method, results, limitations, ownership, approvals and monitoring arrangements. It is also worth recording rejected alternatives. Without that record, a model can later be reused for a purpose that was considered but never evaluated.

Professional responsibility does not end with following an instruction. POPIA sets conditions for processing and safeguarding personal information and for certain solely automated decisions (Protection of Personal Information Act 4 of 2013, sections 8–19 and 71). Professional codes similarly require honesty about limitations, respect for privacy and suitable review. A practitioner needs to raise a concern when data are used unlawfully, evidence is overstated or a system presents an unreasonable risk.

**Read**

**Optional:** Indeed Editorial Team. 2025. 14 Types of Software Development Documentation (Plus Tips)

*There are various types of documentation associated with software development. All of these types are intended for different audiences or have different purposes. This source discusses some of the different types of documentation associated with the software development process.*

## Multiple-choice questions

Select the best answer for each question. The correct answers appear after all five questions.

**1. Why should a data scientist clarify the business decision before collecting data?**

A. To ensure that every available field is retained
B. To identify which data are relevant, lawful and available at the required time
C. To select the most complex algorithm
D. To avoid consulting business specialists

**2. Which situation is the clearest example of information leakage?**

A. A fraud model uses the transaction value available at authorisation
B. A model is tested on records from a later period
C. A loan model uses a repayment outcome that would only be known after the lending decision
D. Missing values are recorded in the data dictionary

**3. What does precision measure in a classification task?**

A. The proportion of all relevant cases that the model identifies
B. The proportion of flagged cases that are genuinely relevant
C. The speed at which a model produces an output
D. The number of variables used by the model

**4. Why is a successful computational notebook not necessarily ready for production?**

A. Notebooks cannot contain machine-learning models
B. Production use also requires controlled code, security, monitoring, support and ownership
C. A notebook cannot be reviewed by another practitioner
D. Production systems may only use novel algorithms

**5. What is the main purpose of segregation of duties in model deployment?**

A. To prevent data scientists from experimenting
B. To ensure that each team uses different programming software
C. To separate incompatible responsibilities and provide appropriate independent review
D. To transfer responsibility for the model to the software vendor

**Correct answers:** 1—B; 2—C; 3—B; 4—B; 5—C.

# Recommended Reading

The following optional readings are not compulsory but are recommended to enrich your understanding of this topic. Full publication details and access links appear in the consolidated References section.

James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. 2023. An Introduction to Statistical Learning with Applications in Python. Cham: Springer. https://hastie.su.domains/ISLP/ISLP_website.pdf.download.html

National Institute of Standards and Technology. 2023. Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1. Gaithersburg, MD: NIST. https://doi.org/10.6028/NIST.AI.100-1

Sandve, G.K., Nekrutenko, A., Taylor, J. and Hovig, E. 2013. “Ten simple rules for reproducible computational research”, PLOS Computational Biology, 9(10), e1003285. https://doi.org/10.1371/journal.pcbi.1003285

# Summary

This topic followed the work of a financial data scientist from the decision and data foundation through to deployment. Tables, messages, text and images can only be combined once their identifiers, timing and business meaning have been reconciled. NLP and computer vision can assist a defined process, but the output still has to be tested in the setting in which it will be used. Where the consequences are material, uncertain cases also require human review.

The work is inherently collaborative. Business, engineering, operations, security, legal, compliance, and risk all have different knowledge and authority. A prototype shows feasibility, not production readiness. The institution still requires controlled code, validation, documentation, monitoring and an identifiable owner before release. Such practices reveal assumptions and limitations and accommodate challenging impacts on customers.

# Self-Assessment Questions

The following questions are not graded. Use them to test your understanding of the key concepts covered in this topic before moving on. Try to answer each question fully in your own words before checking the suggested answers at the back of this study guide.

1. A bank asks a data scientist to build a model using 'all available customer data'. Explain the questions that should be resolved before data are collected and why collecting every field may be inappropriate.
2. An insurance team wants to combine policy tables, adjuster notes and claim photographs. Explain how structured, semi-structured and unstructured data differ, and propose a controlled way to bring these sources together.
3. A prototype fraud model performs well in a notebook. Explain why this result is insufficient for production deployment and identify the additional responsibilities that must be addressed.
4. Develop a measurement plan for a model that prioritises customer complaints. Include at least one technical measure, two business measures and one possible unintended consequence.
5. Explain why documentation and independent challenge form part of technical quality rather than merely administrative compliance in a financial data-science project.

6. A manager asks a junior data scientist to deploy a model despite evidence that it disadvantages a customer group. Explain the professional response that would be appropriate and the stakeholders who should be involved.

# Topic 4
# Big data technologies and challenges

## 4.1 INTRODUCTION

This topic relates to the following module outcome(s):

- Evaluate challenges of working with big data in the field of Finance.

Big data is often introduced through products and platforms. In finance, however, the more useful starting point is the pressure created by data that arrive quickly, differ in form or must be retained at scale. Banks and insurers have always produced records; electronic markets, digital payments and connected channels changed the volume, variety and timing of those records.

The rate at which financial data arrive creates a practical problem. A card authorisation, for example, may require a decision before the transaction completes, even though the data originate in several operational systems. The architecture has to deliver that response without sacrificing accuracy, privacy, security or resilience. Scale alone is of little value if the records are not understood and governed.

This topic explains how these characteristics influence storage, processing and governance. The aim is not to promote a particular platform, but to show why architecture should follow the financial decision, the workload and the controls that apply.

In this topic, you will gain knowledge in the following areas:

- The development and defining characteristics of big data in finance
- Distributed storage, processing, streaming and cloud technologies
- The relationship between big data, AI and machine learning
- Data-quality, privacy, security, lineage and third-party challenges
- Structuring and integrating heterogeneous financial datasets

## 4.2 FROM FINANCIAL RECORDS TO BIG DATA

Money has always been about keeping records. There were ledgers and policies and claims before “big data”, but they were generally bounded and processed in intervals. The rate of

record creation and the scope of information available for analysis were increased by electronic trading, online banking and mobile payments.

In previous analytical systems, selected operational data was often copied into a data warehouse, a curated store that organises consistent, quality-controlled data for reporting and analysis. Warehouses are still useful, but text, images and high-volume event data may need additional storage and processing (we'll come back to data warehouses later in this topic). Cheaper storage, distributed computing and cloud services have made these alternatives affordable.

This is not a simple replacement of old technology by new technology. Banks run their core systems and data warehouses, but add services for streaming, cloud processing or machine learning when justified.

**Read**

**Recommended:** Berman, J.J. 2018. Principles and Practice of Big Data. Elsevier Science: London. Page 1 to 13: Introduction to Big Data

**Optional:** CFI Team. 2020. Big Data in Finance.

*This source sheds a bit more light on the role of big data in finance.*

## 4.3 WHAT MAKES FINANCIAL DATA 'BIG'?

Big data is not defined by a fixed number of rows or terabytes. NIST instead describes several characteristics that may require scalable architecture (Chang & Grady, 2019). The characteristics all start with the letter "V" and are frequently referred to as the Vs of Big Data:

| Characteristic | Meaning | Financial example |
|---|---|---|
| Volume | The amount of data | Several years of transaction history. |
| Variety | Different formats and sources | Tables, market feeds, emails and document images. |
| Velocity | The rate of arrival or required processing | Card authorisations or market events. |
| Variability | Changes in structure, rate or flow | New products or revised message formats. |
| Veracity | The accuracy, completeness and reliability of the data | Checking transaction records for missing values, duplicates or incorrect amounts. |
| Validity | Whether the data are suitable for the intended purpose | Confirming that the selected customer data are appropriate for assessing credit risk. |

| Characteristic | Meaning | Financial example |
|---|---|---|
| Value | Whether using the data produces a worthwhile benefit | Determining whether an additional data source improves a fraud model sufficiently to justify its cost and risk. |

### Real-world example

A card-authorisation stream may be modest in stored size but still create a big-data problem when thousands of events must be assessed within milliseconds. In this case, velocity and the required response time matter more than the number of terabytes retained.

Read

**Supporting:** Chang, W.L. and Grady, N. (2019), NIST Big Data Interoperability Framework: Volume 1, sections 3.1–3.2, pp. 9–12.

**Supporting:** Mlacki, J. 2026. 7 Vs of Big Data: What They Are & Why They Matter.

## 4.4 BIG-DATA TECHNOLOGIES IN FINANCE

The characteristics above describe the pressures placed on an analytical environment. Big-data technologies are possible responses to those pressures, rather than an end in themselves.

Big-data technologies distribute storage or processing across several computing resources. Hadoop popularised distributed storage and parallel batch processing, although distribution also introduces coordination, security and consistency problems. NIST's architecture discussion places these capabilities within a broader set of scalable storage, processing, cloud and analytical services (Chang & Grady, 2019, sections 4.1–4.4).

Apache Spark supports distributed Structured Query Language (SQL), batch, streaming and machine-learning workloads. Apache Kafka stores and routes continuing event streams. In finance, Spark might process several years of transactions, whilst Kafka or a comparable service carries card events to a fraud-monitoring process. These products illustrate different capabilities; they are not a required product list.

Enterprise and managed offerings package these capabilities into supported services. Databricks provides a managed environment built around Apache Spark and lakehouse architecture, while Google BigQuery provides a managed cloud data warehouse for large analytical workloads. They illustrate common industry options rather than a prescribed product choice.

Cloud platforms provide comparable managed services. A data lake retains varied formats, a warehouse presents curated structured data, and a lakehouse attempts to combine flexible storage with warehouse controls. Since these labels are not perfectly standardised, the important questions concern storage, cataloguing, processing, security and access. The main distinctions are summarised below.

| Architecture | Advantages | Limitations |
|---|---|---|
| Data warehouse | Consistent, curated data for reporting and repeatable analysis. | Less flexible for raw, rapidly changing or unstructured data. |
| Data lake | Stores large volumes in varied formats and supports later reuse. | Can become difficult to interpret or govern without strong metadata and controls. |
| Lakehouse | Combines flexible storage with selected warehouse-style controls. | Adds architectural complexity and does not remove the need for governance. |

**Read**

**Supporting:** IBM.com. n.d. Data warehouses versus data lakes versus data lakehouses.

*This source allows you to delve a bit deeper into the differences between these 3 data storage approaches.*

## 4.5 BATCH, STREAMING AND ANALYTICAL WORKLOADS

A batch process handles a bounded collection at an interval, such as end-of-day positions. Stream processing handles events as they arrive, for example before a payment is authorised.

### Did you know?

Real-time payment does not remove batch processing. A transaction may be screened and confirmed immediately, yet the institution will still reconcile totals, investigate exceptions and prepare settlement or regulatory reports later.

Batch and streaming designs are often combined. For example, a card event can be assessed immediately and can then be retained for reconciliation, reporting and later model development. They don't require the same controls. A stream contends with events arriving late, twice or out of order, while a batch process must contend with cut-off times, incomplete files and the consequences of rerunning a reporting period. Either way, the institution needs traceable processing and a way to investigate failures.

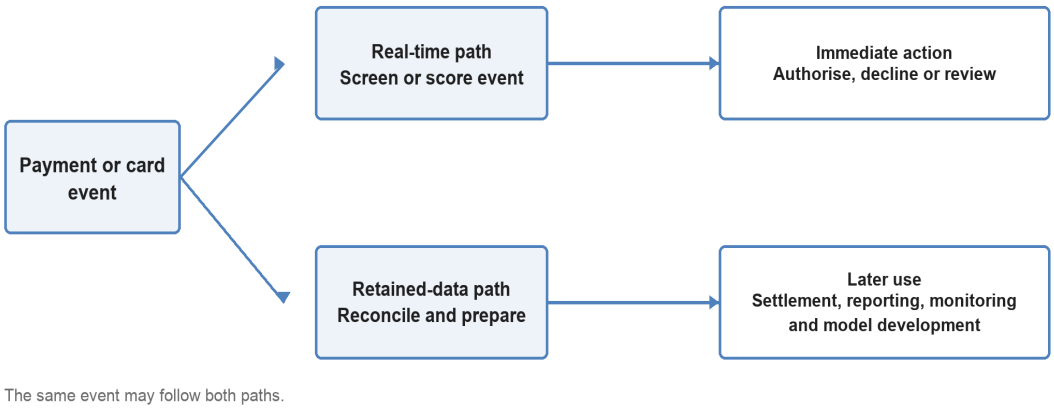



**Figure 4.1: Real-time and retained-data processing paths**

*Source: Author*

Workloads differ in their practical requirements. Interactive queries prioritise response time, scheduled reports prioritise repeatability, and model training may require substantial resources only at intervals. Production scoring may need low latency, or little delay, and high availability, meaning that the service remains accessible when required.

**Read**

**Supporting:** Chang, W.L., Boyd, D. and Levin, O. (2019), NIST Big Data Interoperability Framework: Volume 6, sections 4.1.2–4.1.5, pp. 23–28. Focus on collection, preparation, analytics, processing, the data lifecycle and security rather than on a particular vendor product.

**Optional:** Lu, T. 2024. Batch vs Stream Processing: When to Use Each and Why It Matters.

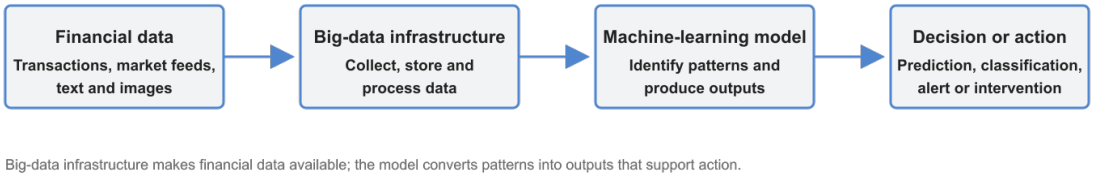
*This optional source sheds a bit more light on the difference between batch and stream processing.*

## 4.6 AI AND MACHINE LEARNING IN THE BIG-DATA ENVIRONMENT

Artificial intelligence (AI) refers broadly to systems that perform tasks associated with human intelligence, such as interpreting language, recognising patterns or supporting decisions. Machine learning (ML) is a branch of AI in which models learn patterns from data rather than relying only on rules specified in advance.

Big-data technologies support this work by collecting, storing and processing information that may be too large, varied or fast-moving for conventional tools. Machine-learning models can then use these data to produce predictions, classifications or alerts. In finance, possible applications include identifying unusual card transactions, estimating credit risk, extracting information from company reports and analysing market activity. The relationship can be understood as a sequence, as depicted in Figure 4.2.

A method that works on a small extract can become too slow, or no longer fit into memory, when it is applied to hundreds of millions of records. This is an engineering problem as well as an analytical one. Sampling, partitions, mini-batches, multiprocessing or distributed computation are possible responses, and in some cases an incremental or approximate algorithm is more practical. Whatever choice is made, it still has to be checked on representative held-out data; faster computation is not a substitute for a sound result.




Big-data infrastructure makes financial data available; the model converts patterns into outputs that support action.

**Figure 4.2: From data to action through big data and machine learning**

*Source: Author*

Each component has a different role. The infrastructure makes the data available for analysis, while the model estimates patterns that may support a financial decision. The resulting output may, for example, flag a payment for investigation, contribute to a lending assessment or generate a market-risk indicator.

More observations do not automatically produce a better model. Financial relationships change, records through time are often dependent, and historical data can carry earlier bias. Incorrect labels, a poorly chosen target or information from after the decision can also create impressive but misleading performance. The last of these problems is known as data leakage.

Statistical performance on its own is not enough to evaluate an AI or ML system. Cao (2023) describes financial data science as an interaction between data, business processes, behaviour and external conditions. The model therefore has to be judged at the point where its output is used: does it support the intended decision, does it remain dependable as conditions change, and what happens when it is wrong? In a big-data environment, those questions extend to data quality, model risk, cybersecurity and dependence on external technology providers.

**Read**

**Optional:** OECD. 2021, Artificial Intelligence, Machine Learning and Big Data in Finance: Opportunities, Challenges, and Implications for Policy Makers. Focus on Sections 1 and 2.

*This is quite a lengthy, but interesting optional read that discusses the intersection of AI, ML and Big Data in the world of finance.*

Optional: Wang, Lidong & Alexander, Cheryl. (2016). Machine Learning in Big Data. International Journal of Mathematical, Engineering and Management Sciences. 1. 52-61. 10.33889/IJMEMS.2016.1.2-006. https://www.researchgate.net/publication/327386452_Machine_Learning_in_Big_Data

*This paper presents a very good overview of the application (and types) of machine learning in Big Data, some examples and challenges.*

## 4.7 CHALLENGES AROUND FINANCIAL BIG DATA

The data to action sequence shows how infrastructure can help machine learning. It also highlights several places where bad data, weak controls or outside dependencies can jeopardise the outcome.

Financial institutions were designed for other purposes. A customer may have multiple identifiers. An authorisation record, clearing record and settlement record represent different stages of a transaction. External market data comes with their own calendars, corrections, licensing terms. Without the business meaning, the data set can appear complete when these records are combined but still be false.

Missing or duplicated records are obvious quality problems, but labels and definitions can be just as troublesome. For example a 'fraud' label might only contain the cases that were investigated and confirmed. A migration might change the meaning of a field without changing the name of the field. The proposed use determines which checks matter, and important totals should be reconciled to a trusted source where one exists.

Scale brings with it operational and governance risks. More distributed components mean a larger security perimeter, more copies for analysis can subvert access control, and reliance on a small core of specialist providers creates concentration risk.

**Read**

**Optional:** OECD. 2021, Artificial Intelligence, Machine Learning and Big Data in Finance: Opportunities, Challenges, and Implications for Policy Makers. Focus on Section 3.

*This is quite a lengthy, but interesting optional read that discusses the intersection of AI, ML and Big Data in the world of finance.*

## 4.8 STRUCTURING BIG DATA FOR FINANCIAL USE

Scale alone cannot address the above challenges. They require intentional decisions about how data are described, organised and maintained.

The first step in structuring is to create metadata that defines the meaning, format, ownership, origin and permissible use of data. Catalogues make data sets discoverable and dictionaries define fields and business terms. Schema-on-read imposes structure at the time data is accessed, while schema-on-write imposes structure before data is stored. Many platforms use both.

The storage format also affects the ability to use large data sets. Apache Parquet is a column orientated file format that stores schema information allowing analytical systems to read only selected columns without scanning all the fields. In Python, data frame libraries like Polars may use columnar and parallel processing to work with data stored in formats such as Parquet efficiently. These tools improve performance, but the institution must provide field definitions and governance.

Reference data provides controlled values such as currency codes and instrument classifications while master data seeks consistent records for customers, products or counterparties. Why it matters: Because things change, effective dates matter. Thus, a transaction must be associated with the version that was valid at the time of the transaction.

### Real-world example

A customer may have different identifiers in a current-account system, a credit-card platform and a mobile-banking application. Combining those records requires governed matching rules; simply joining on a similar name could merge two different people or leave one customer represented several times.

Partitioning divides a dataset into manageable parts, perhaps by transaction date, whilst an index helps locate selected records. Efficiency should not remove analytical meaning: aggregation may

discard evidence needed for fraud investigation, and linked attributes may permit re-identification even after direct identifiers are removed.

**Read**

**Recommended:** Berman, J.J. 2018. Principles and Practice of Big Data. Elsevier Science: London. Page 155 to 167: Standards and Data Integration

**Visit**

The following sites provide an overview of a few Python libraries which may be handy when working with financial data (or in data science in general):

https://corporatefinanceinstitute.com/resources/data-science/python-libraries-for-data-analysis/

https://www.efinancialcareers.co.za/news/python-libraries-for-finance

https://www.datacamp.com/blog/top-python-libraries-for-data-science

## 4.9 INTEGRATING DIFFERENT FINANCIAL DATASETS

Data integration is the combining of sources for a common purpose. Extract-transform-load (ETL) transforms data before loading, while extract-load-transform (ELT) transforms data in the target platform. An application programming interface (API) supports defined requests between systems and event streams deliver ongoing updates. Design matters: latency, control, reproducibility.

Integration is more than a join of tables. Before the outcome can be trusted, identifiers, units, currencies, time zones, calendars, levels of detail and business definitions must be reconciled. Entity resolution then asks if records from different sources describe the same customer, organization or instrument. A false match links unrelated records, and a missed match splits one entity into multiple identities. Successful software execution does not answer the question of the matching rules nor that of the uncertain cases; they are still to be tested and recorded.

End-to-end lineage captures the source of integrated data, the rules applied and the reports or models that depend on them. Then you can track an error back to affected outputs. "Given ongoing concerns about legacy systems and fragmented data estates, integration

should be judged on whether the resulting data is accurate, timely and explainable and fit for the financial decision.

### Activity: Integrating Financial Data with Python

This activity requires you to revisit some of your existing Python knowledge to work with financial data. You are provided with the two following spreadsheets, containing related financial data:

customers.csv
transactions.csv

*These files are available for download in the course page.*

Using Python (in any IDE) and Pandas:

(1) inspect the columns, data types and missing or invalid values;
(2) standardise the customer identifiers and transaction dates;
(3) convert amount to a numeric field while retaining evidence of invalid values;
(4) join the datasets without discarding unmatched records; and
(5) identify duplicate or unmatched records. Display the resulting dataset.

*A sample solution to the activity is available for download in the course page.*

## Recommended Reading

The following optional readings are not compulsory but are recommended to enrich your understanding of this topic. Full publication details and access links appear in the consolidated References section.

Cao, L. 2023. "AI in finance: Challenges, techniques, and opportunities", ACM Computing Surveys, 55(3), Article 64, pp. 1–38. https://doi.org/10.1145/3502289. Free preprint: https://arxiv.org/pdf/2107.09051

Berman, J.J. 2018. Principles and Practice of Big Data: Preparing, Sharing, and Analyzing Complex Information. 2nd ed. London: Academic Press. Read Chapter 1, pp. 1–13, and Chapter 7, pp. 155–167.

López de Prado, M. (2018), Advances in Financial Machine Learning. (It's not necessary to review this entire text, but some of the sections will be interesting and align with the discussions in this topic) https://www.proquest.com/legacydocview/EBC/5240570?accountid=198890

## Summary

Big data is best understood here as a set of pressures created by volume, variety, velocity or variability, not as a fixed dataset size. Distributed storage, processing and event-streaming tools respond to those pressures, while warehouses, lakes and lakehouses organise data for different uses.

Batch and streaming processes commonly operate together because an immediate financial decision still has to be reconciled, reported and investigated later.

Scale does not repair a poor analytical design. A large platform can still contain an unsuitable target, biased labels or records whose meaning has been lost. Financial use depends on metadata, consistent definitions, reference and master data, security, quality controls and end-to-end lineage. The useful measure of integration is therefore whether the data are accurate, timely, explainable and fit for the decision—not how much data the institution managed to store.

## Self-Assessment Questions

The following questions are not graded. Use them to test your understanding of the key concepts covered in this topic before moving on. Try to answer each question fully in your own words before checking the suggested answers at the back of this study guide.

1. A dataset contains only two million card transactions but must be assessed within milliseconds and arrives continuously. Explain why it may still present a big-data problem even if its stored volume is modest.
2. Compare a batch architecture with a streaming architecture for fraud detection. Explain which parts of the process may use each approach and why both might be retained.
3. A financial institution proposes moving all raw data into a cloud data lake. Evaluate the proposal in terms of value, data quality, privacy, security, lineage and provider dependency.
4. Explain why a larger training dataset does not necessarily produce a better financial machine-learning model. Use examples involving time, labels or changing business processes.
5. Two systems contain different identifiers and balances for what appears to be the same customer account. Describe the integration work required before the records can be used together.
6. Design a high-level data path for a real-time card-fraud system, from the transaction event to later monitoring and reporting. Identify where streaming, scalable processing, structured storage and governance controls are required.

# Topic 5
# Techniques to identify key problems

## 5.1 INTRODUCTION

This topic relates to the following module outcome(s):

- Use suitable techniques to identify key problems and select relevant approaches to finding solutions, given cases from the finance industry

A data-science project should begin with a decision, not with an algorithm. A bank may want to reduce losses, an insurer may want to route claims more efficiently, or an asset manager may want to rebalance a portfolio. These broad aims still have to be converted into a precise problem: who or what is being assessed, what outcome matters, when the estimate is required and what action will follow.

This topic follows that path from an initial concern to a measurable intervention. It considers practical problem-framing techniques, the examination and organisation of data, common causes of unreliable evidence, and the distinction between predicting an outcome and choosing an action. The emphasis is not on a particular modelling technique. It is on deciding whether analysis is needed, whether the available evidence is fit for purpose and how the result will improve a financial process.

In this topic, you will gain knowledge in the following areas:
1. Identifying and framing a financial problem
2. Examining, organising and storing relevant data
3. Responding to inaccurate or uninterpretable data
4. Predicting and optimising business outcomes
5. Using data to support an action

## 5.2 BEGIN WITH THE DECISION

A useful problem statement connects a current condition to a decision that someone can take. “Customer attrition is increasing” describes a concern. “Which customers should the retention team contact this month, and which offer should be made?” identifies a decision. The second statement also exposes the information that will be needed: the unit of analysis is a customer, the decision date is known, and the proposed action has a cost.

A short framing discussion should establish the stakeholder, the desired outcome, the constraints and the consequences of error. Interviews and process observation are useful because the written

request may omit practical details. A process map can then show where data are produced, where a decision is made and where delay or rework occurs. There are various ways in which a problem may be solved. One such approach, The "five whys" technique, can be used to trace a visible problem towards possible underlying causes by repeatedly asking why it occurred (Serrat, 2009). The result depends on the questions and available evidence, so it should guide further investigation rather than be treated as proof of causation. The following table presents a few sample questions you could ask.

| Question | What it establishes | Illustrative lending example |
|---|---|---|
| What decision will change? | The practical purpose | Whether an application is approved, declined or referred |
| Who or what is assessed? | The unit of analysis | One application at a specified decision time |
| What outcome matters? | The target and horizon | Material arrears within twelve months |
| What errors matter most? | The cost of mistakes | Declining a reliable borrower or approving an unaffordable loan |
| What constraints apply? | Legal and operational boundaries | Fairness, affordability rules, response time and review capacity |
| How will improvement be measured? | The success criterion | Loss, approval quality, customer impact and processing time |

### Did you know?

A model can answer the wrong question very accurately. If the target is "accounts sent to collections", the model may learn past collection policy rather than underlying repayment difficulty. Target definitions should therefore be checked against the decision the institution actually wants to improve.

## 5.3 FROM A BUSINESS QUESTION TO AN ANALYTICAL QUESTION

A business question should be converted into something that can be investigated. A common practical classification separates analytical questions into four forms: descriptive asks what happened, diagnostic asks why it happened, predictive estimates what may happen, and prescriptive considers what should be done (IBM, n.d.). These questions can occur in sequence, but a prediction is not always needed. If two reports disagree because a process is broken, reconciliation and repair may solve the business problem more directly than a model. The following table lists the four question types and their purpose:

| Question type | Purpose | Example |
|---|---|---|
| Descriptive | Summarise observed events | How many card disputes were received by product and week? |
| Diagnostic | Investigate plausible causes | Which process changes coincide with the rise in disputes? |
| Predictive | Estimate an outcome not yet known | Which current transactions are most likely to become confirmed fraud? |
| Prescriptive | Compare feasible actions | Which review threshold gives the best balance between prevented loss and customer disruption? |

Before modelling begins, the team should write a target definition and an observation point. The target states what counts as the outcome; the observation point states when the available inputs are measured. This prevents information from the future entering the analysis. A baseline should also be recorded, whether it is the current rule, a simple average or the existing operational result. For example, suppose the current referral rule identifies 70% of loans that later enter serious arrears, incorrectly refers 8% of loans that remain current and requires two hours of review per 100 applications. A proposed model could be required to identify at least 75% of later arrears without increasing the 8% false-positive rate, while reducing review time. The figures are illustrative; their purpose is to make the claimed improvement testable.

**Read**

**Supporting:** Xie, T. 2024. How to Answer Business Questions with Data.

*This source sheds some light on moving from a business question to the data analysis process.*

## 5.4 EXAMINE THE DATA BEFORE SELECTING A METHOD

A defined question narrows the search for evidence: it identifies the records, dates and units that ought to matter. The next task is to check whether those data exist and represent what the team assumes.

Initial data examination tests whether the proposed analytical question is answerable. It begins with meaning rather than calculation. The analyst should identify the source and owner of each field, the time period covered, the unit represented by one row, the process that generated the record and any later corrections. A field called “balance”, for example, may mean ledger balance, available balance or balance at a different reporting time.

A data profile then describes types, ranges, missing values, repeated records, category frequencies and changes over time. Visualisations may reveal skewed distributions, step changes or unlikely clusters, but an unusual value is not automatically an error. A large transaction may be genuine and important. Its meaning should be checked against the source process before it is removed or altered. The following table represents a few concepts (and associated questions to ask) when looking at financial data for analysis.

| Check | Question | Possible response |
|---|---|---|
| Coverage | Does the period include the customers, products and conditions relevant to the decision? | Restrict the claim, obtain further data or redesign the question |
| Completeness | Which values or records are absent, and why? | Investigate the missingness and document any treatment |

| | | |
|---|---|---|
| Consistency | Are units, codes and definitions stable across sources and time? | Standardise carefully and retain the original values |
| Timeliness | Would each input be available at the decision point? | Remove future information or redesign the data extract |
| Representativeness | Who or what is under-represented? | Use appropriate sampling, evaluation and limitations |
| Interpretability | Can the field be explained and reproduced? | Trace lineage, consult the owner or exclude it |

These checks also constrain method selection. The target, data structure, timing, sample size and need for explanation narrow the methods that are suitable. Complex temporal data, for example, may require sequence-aware methods or carefully constructed time-based features; a simple logistic regression fitted without representing that structure could miss an important pattern. Data quality still sets the limit: a more complex method cannot repair unreliable inputs.

Read

**Supporting:** Datafold. n.d.. What is data quality?

*This source delves a bit deeper into various types of data quality considerations.*

## 5.5 ORGANISE AND STORE DATA FOR THE PROBLEM

Organisation should follow the intended use without destroying traceability. Raw source data are normally retained separately from cleaned or derived data. Stable identifiers and timestamps permit records to be joined and ordered, whilst a data dictionary records definitions, formats, permitted values and ownership. The analytical dataset should be reproducible from documented transformations rather than assembled through unrecorded manual changes.

Storage follows the structure, scale, access pattern and sensitivity of the data: consistent account records may fit a relational table, while documents or large analytical files are often better suited to object storage. The newest technology is not automatically the best one. The design still has to preserve meaning, security, lineage and usable performance, with access limited to the stated purpose. Personal information should not be copied simply in case it becomes useful later.

One way of organising a lakehouse is the medallion design. The bronze layer retains raw ingested data, the silver layer contains cleaned and validated data, and the gold layer contains curated datasets prepared for a defined analytical or reporting purpose. Routine work may begin with gold data, but a new question can require controlled access to the more detailed silver or bronze records. Moving down the layers also increases the need to understand quality, sensitivity and lineage (Databricks, 2026).

### Real-world example

Keep the “as known then” version of the evidence. Financial records are often corrected after an event. Reconstructing a model with the latest corrected table may introduce information that was unavailable at the original decision time. Dated snapshots or effective-date fields help preserve a defensible historical view.

Read

**Supporting:** DolphinDB. 2025. Best Practices for Financial Data Storage. https://medium.com/@DolphinDB_Inc/best-practices-for-financial-data-storage-d05dc7529568

*This source sheds some light on the best practices for storing financial data. It’s written from the perspective of a specific platform, but these same considerations could apply to other similar storage platforms.*

## 5.6 INACCURATE AND UNINTERPRETABLE DATA

An inaccurate value and an uninterpretable value are not quite the same problem. Capture errors, duplicated events, stale reference data, failed interfaces and incorrect transformations alter the record itself. A value can also be recorded exactly as produced while its meaning remains uncertain. Zero income, for example, could mean no income, no response, not applicable or a system default; treating these cases as identical creates false precision.

The response depends on the cause and scale of the problem. An analyst may correct a value from an authoritative source, apply a documented derivation, mark it as missing, exclude the affected observations or stop the analysis. Imputation can support some analyses, but it estimates a missing value; it does not recover evidence that was never recorded. The treatment and its effect on the result should be tested and disclosed.

Even an understandable input can be difficult to interpret. A confounding variable is related to both the input being studied and the outcome. For example, a hardship programme may be offered mainly to already distressed borrowers and may also reduce later arrears. If the programme is ignored, an analyst could attribute its effect to another borrower characteristic. A strong association therefore does not establish a causal explanation.

Uninterpretable inputs also weaken accountability. If neither the analyst nor the data owner can explain how a feature was produced, it may be impossible to challenge the model or reproduce the

decision. A technically strong association is not enough reason to retain such a variable in a material financial process.

Read

**Supporting:** Gomstyn, A. and Jonker, A. 2025. Data quality issues and challenges. https://www.ibm.com/think/insights/data-quality-issues

*Review this source to learn a bit more about the issues related to data quality and how to address them.*

## 5.7 PREDICTION AND OPTIMISATION

Only after the evidence and its limitations are understood should the team decide whether prediction, optimisation or a simpler intervention is appropriate.

Prediction estimates an unknown outcome, such as the probability of default or expected claim cost. A realised classification may later be marked correct or incorrect, while a probability estimate is evaluated across many cases. Optimisation uses estimates and other information to select the best feasible action for a stated objective and constraints, such as allocating a review budget or choosing a portfolio under risk limits. It may have a mathematical optimum under those assumptions, but not a universally "perfect" answer: changing capacity, costs or risk limits can change the preferred action. As with many real-world scenarios, perfection is a lofty goal, which may be difficult (and in some cases impossible) to attain; so, optimisation leads to a result as close as the approach can come to *'perfect,'* given the specific input data and time frame in which to find a solution. An accurate risk estimate therefore does not by itself determine how much credit should be offered.

An objective needs to state more than the preferred direction of change. For instance, *'Maximise approvals'* can increase volume while weakening affordability or loss performance, and '*Minimise fraud loss'* can block large numbers of genuine transactions. A workable objective balances the relevant measures and fixes limits that cannot be exceeded. It also needs stress or sensitivity tests because the financial conditions surrounding the decision will change.

Read

**Prescribed:** Aldridge, I. and Avellaneda, M. (2021), Chapter 2, pp. 27–35.

*Read the discussion of training, cross-validation, out-of-sample testing, overfitting and the limits of a financial prediction. It's not necessary to spend any time on the mathematics/derivations presented in this part of the text.*

## 5.8 USING DATA TO DRIVE ACTION

The difference between prediction and optimisation makes clear what is produced as output. The output still needs a separate operational design before it can have an impact on a customer, transaction or portfolio.

An analytical output is only of value if it feeds into a workable decision process. A score can be used to rank cases, but a policy still dictates what occurs at various score levels. The policy must consider capacity, cost and risk appetite. The question is not whether a model predicts fraud, but whether the top 500 cases improve outcomes over the current queue if a fraud team can only look at 500 alerts a day.

Evidence reaches practice through an escalation route, an owner and a decision rule. For instance, a low impact action can be automated when the controls are strong, but a material or unusual case may require human review. The result must then be fed back into the analytical process. Without that feed-back the institution cannot know if the intervention worked or if behaviour changed after its introduction.

## Multiple-choice questions

Select the best answer for each question. The correct answers appear after all four questions.

**1. Which statement is the clearest decision-centred problem?**

A. The institution has a large customer dataset
B. Which applications should be approved, referred or declined at the decision point?
C. The analytics team wants to test a new algorithm
D. Loan losses have changed over time

**2. Why should an observation point be defined before a model is developed?**

A. To guarantee that the model will be accurate
B. To determine which information would actually be available when the decision is made
C. To ensure that every field is retained
D. To remove the need for a baseline

**3. A dataset records all missing income values as zero. What should the analyst do first?**

A. Treat every zero as confirmed evidence of no income
B. Replace every zero with the average income
C. Investigate how the field was captured and what zero represents in the source system
D. Remove the income field without consulting its owner

**4. Which example describes optimisation rather than prediction?**

A. Estimating the probability that a transaction is fraudulent
B. Forecasting next month's claim volume
C. Ranking borrowers by estimated probability of default
D. Selecting which fraud alerts to review within the available investigator hours

**Correct answers:** 1—B; 2—B; 3—C; 4—D.

# Recommended Reading

The following optional readings are not compulsory but are recommended to enrich your understanding of this topic. Full publication details and access links appear in the consolidated References section.

Basel Committee on Banking Supervision. 2013. Principles for Effective Risk Data Aggregation and Risk Reporting (BCBS 239). Basel: Bank for International Settlements. https://www.bis.org/publ/bcbs239.pdf Focus on Principle 2, pp. 6–7, and Principles 3–6, pp. 8–10.

Aldridge, I. and Avellaneda, M. (2021), Big Data Science in Finance. Read Chapter 2, pp. 27–35 only.

# Summary

A useful data-science problem starts with a decision and a defined outcome, not with a preferred algorithm. Once the business concern has been expressed as a descriptive, diagnostic, predictive or prescriptive question, a practical measure and baseline allow the team to test whether analysis adds any value.

The data also have to be read in the context of the process that produced them. Accuracy, meaning, coverage, timing and representativeness limit what can be concluded. Storage and preparation

should preserve lineage and history. Prediction estimates an outcome; optimisation selects an action under objectives and constraints. Both are ultimately judged by what changes in the financial process and for the people affected by it.

## Self-Assessment Questions

The following questions are not graded. Use them to test whether you can move from a broad concern to a defensible data-driven action.

1. Rewrite the statement *"loan losses are too high"* as a decision-centred problem statement. Identify the unit of analysis, outcome, observation point and possible action.
2. Explain why a predictive model may not be necessary to solve a data-related business problem. Give one financial example.
3. A dataset records missing income as zero. Explain why this creates both an accuracy and an interpretation problem, and describe an appropriate investigation.
4. Distinguish prediction from optimisation using a fraud-review example.
5. Describe the evidence needed to show that a new analytical intervention improved the business outcome rather than merely producing an accurate model.

# Topic 6
# Practical cases of specific data scientist tasks in the Finance industry

### 6.1 INTRODUCTION

This topic relates to the following module outcome(s):

- Demonstrate knowledge of the structures, regulation and key systems/data vendors of the Finance industry.
- Demonstrate an understanding of different Finance areas relevant to data science.
- Demonstrate an understanding of general responsibilities of a data scientist in the Finance sector.
- Evaluate challenges of working with big data in the field of Finance.
- Use suitable techniques to identify key problems and select relevant approaches to finding solutions, given cases from the finance industry

The cases in this topic bring together the earlier discussion of financial data, responsibilities, infrastructure and problem framing. They are not recipes. Each case was developed for a particular institution, dataset and decision, and its result cannot be transferred without further evidence.

Three cases are used. Project Aurora examines collaborative detection of money-laundering networks; a World Bank project in Ethiopia examines psychometric information as an aid to lending

decisions; and the prescribed textbook examines clustering for portfolio construction. Together they show that practical data science includes the definition of a problem, the construction of suitable evidence, evaluation against a baseline and controls around the resulting action.

| Case | Financial task | Main lesson |
|---|---|---|
| Project Aurora | Risk analysis and suspicious-activity detection | Network-wide evidence can reveal patterns hidden from institutions acting separately |
| Ethiopian psychometric scoring | Credit assessment and customer access | An innovative score still requires operational testing, fairness review and a fallback process |
| Portfolio clustering | Investment grouping and portfolio construction | Unsupervised groups require financial interpretation and out-of-sample evaluation |

## 6.2 A FRAMEWORK FOR READING A CASE

A case is most useful when it is read as a chain of evidence. Start with the decision and the current baseline, then establish how the data were generated, which method was applied and how the result was tested. Only after that should the reader ask what controls would be needed in live use. An improvement reported in a case is not enough by itself; the comparison, measure and testing conditions have to fit the intended decision. The following are a few elements to consider when reviewing a case study:

| Element | Question to ask |
|---|---|
| Problem | What decision or process is being improved, and for whom? |
| Data | What do the records represent, what is missing and what may be sensitive? |
| Method | Why is this method suited to the pattern or decision? |
| Evaluation | What baseline and measures were used, and was testing genuinely out of sample? |
| Operation | How does an output become an action, review or escalation? |
| Risk | What could fail, who may be affected and how would the problem be detected? |

## 6.3 CASE 1: COLLABORATIVE DETECTION OF FINANCIAL CRIME

The framework can now be applied to three very different cases. The first concerns a problem in which relevant patterns may extend beyond the data held by one institution.

Project Aurora was a proof of concept led by the BIS Innovation Hub. It addressed a limitation of conventional anti-money-laundering monitoring: each institution sees only part of a payment network, while criminal activity may span several institutions and jurisdictions. The project generated synthetic payment data containing known laundering patterns and compared monitoring at institutional, national and cross-border levels.

Aurora compared a rule-based benchmark with logistic regression, an artificial neural network, a supervised graph neural network and an unsupervised isolation forest. The graph neural network used links among accounts and transactions to

identify relationships across the payment network, while isolation forest was used to identify unusual patterns and anomalies. The project reported recall, precision and F1: recall indicates how much known suspicious activity was found, precision indicates how many alerts were relevant, and F1 summarises the balance between the two where suspicious cases are rare (BIS Innovation Hub, 2023).

In the synthetic experiment, rule-based monitoring detected up to 25% of money launderers in the siloed scenario. The machine-learning models detected more than twice as many, while the graph neural network detected approximately 80% when it used the cross-border view. These results are promising, but they are not a deployment decision. Every local implementation would still require specification and validation, and cross-institution analysis introduces privacy, security, legal and governance questions. The case shows that the boundary of the available data can matter as much as the algorithm (BIS Innovation Hub, 2023).

### Did you know?

Fraud detection and money-laundering detection are related but not identical. Fraud controls commonly protect a customer or institution from unauthorised loss. Anti-money-laundering controls seek suspicious patterns that may indicate the movement of criminal proceeds. The labels, legal duties, time horizons and escalation routes therefore differ.

Read

**Supporting:** BIS Innovation Hub (2023), Project Aurora, Executive summary, pp. 10–15; section 4.3, pp. 39–46; and Chapter 5, pp. 65–73. Focus on the problem, machine-learning experiment and metrics, followed by the data, explainability, privacy and implementation cautions.

**Further reading:** Zhou et al. (2020), pp. 57–64. This open-access review introduces graph-structured data, message passing and the main GNN model families; the detailed derivations are optional.

## 6.4 CASE 2: PSYCHOMETRIC CREDIT SCORING IN ETHIOPIA

Project Aurora stretched the boundary of available transaction data. The second is the converse problem, a lending decision where there is no conventional financial evidence.

A World Bank project explored whether psychometric data could support women entrepreneurs who lacked the collateral or conventional credit history required for a business loan. Applicants completed a computer-based assessment designed to capture characteristics associated with entrepreneurial ability and

willingness to repay. The score was intended to strengthen a thin financial history, not to guarantee repayment.

The report does not name the proprietary algorithm used for the EFL score, but it does describe its evaluation. In the pilot, 2,496 borrowers completed the assessment and 1,132 loans were disbursed under the lender's normal criteria. A regression using 818 observations found that the score remained statistically significant after accounting for age and years in business. Higher-scoring customers were reported to be seven times as likely to repay. Over the following 17 months, the proportion of customers who became more than 30 days overdue at any point fell from 7% in the lowest displayed score band to 1% in the highest. The score still had to fit the application process, be understood by staff and be checked against subsequent repayment outcomes (Alibhai et al., 2018).

But the decision to lend is one that carries material risks. Language, education, disability or familiarity with a digital interface may affect the test without measuring creditworthiness. Applicants require a transparent process, a way to review or provide alternative evidence, and the protection of their personal information. The case demonstrates the potential inclusion benefit of alternative data, but also its limits.

Read

**Supporting:** Alibhai, S. et al. (2018), Disruptive Finance: Using Psychometrics to Overcome Collateral Constraints in Ethiopia. Read pp. 1–8 for the motivation, pilot design and early results, then pp. 9–13 for scaling problems, market reaction and the conclusions. The official report is 13 pages long.

## 6.5 CASE 3: CLUSTERING FINANCIAL INSTRUMENTS

The first two cases concern supervised decisions supported by known outcomes. The third case turns to an unsupervised method, where no predefined correct grouping is available.

Aldridge and Avellaneda use clustering to organise financial instruments with similar return behaviour and then form portfolios from the resulting groups. For K-means, the analyst has to choose the number of clusters (k), starting centres and a stopping condition. The discussed formulation minimises squared Euclidean distance to the cluster centroids. Its Python illustration uses five clusters, K-means++ initialisation, a maximum of 500 iterations, ten initialisations and random_state=0. These are modelling choices rather than neutral defaults: different values of k or different starting points may produce different groups. The chapter therefore tests several cluster counts and compares the resulting portfolios out of sample. Unlike supervised prediction, there is no labelled "correct" portfolio (Aldridge & Avellaneda, 2021, pp. 264–265, 275–295 and 311).

Without a labelled correct answer, evaluation changes. A mathematically compact cluster is not automatically a useful investment group. Its composition, stability and financial meaning have to be examined, after which the resulting portfolio can be tested on later data. The textbook illustrates the process with cryptocurrencies, commodities and equities and reports both in-sample and out-of-sample results.

This is an analytical example, not an investment recommendation. Returns and correlations change, while transaction costs and liquidity affect what can be implemented. Repeated choices about the sample, dates, features and number of clusters can also overfit the historical period. A comparison with a simple portfolio on later data is therefore essential.

Read

**Prescribed:** Aldridge, I. and Avellaneda, M. (2021), Chapter 9, pp. 262–265 for the purpose of clustering and the K-means choices; pp. 275–295 for the portfolio application and out-of-sample comparison; and Appendix 9.A, p. 311 for the Python settings.

*The matrix derivations and detailed spectral-clustering equations are not required.*

## 6.6 COMPARING THE CASES

Across all three cases, it's clear that the method is only one part of the solution. Aurora depends on lawful collaboration, the Ethiopian case on a fair and workable application process, and portfolio clustering on financial interpretation and disciplined testing. To be successful, a pilot still needs local

validation, controlled implementation and monitoring, and a clear decision about when human review is required. The following table represents a summary of the three cases across a few variables:

| Dimension | Aurora | Psychometric scoring | Portfolio clustering |
|---|---|---|---|
| Decision | Which activity warrants investigation? | Which applicant may proceed or requires further evidence? | How might instruments be grouped for portfolio construction? |
| Evidence | Payment networks with embedded suspicious patterns | Assessment responses and later repayment outcomes | Historical returns and relationships among instruments |
| Baseline | Institution-level rules | Existing collateral- and history-based process | A simple portfolio or established allocation method |
| Key risk | Privacy and transfer from synthetic to live data | Unfair exclusion or proxy effects | Unstable clusters and historical overfitting |
| Human role | Investigation and escalation | Explanation, review and lending judgement | Investment interpretation and mandate oversight |

## Multiple-choice questions

Select the best answer for each question. The correct answers appear after all four questions.

**1. Why did Project Aurora examine wider national and cross-border payment views?**

A. To avoid using transaction data
B. To identify relationships that may be hidden when each institution analyses its data separately
C. To guarantee that every alert represents money laundering
D. To remove the need for privacy controls

**2. Why are precision and recall both relevant in financial-crime monitoring?**

A. They measure only how quickly a model runs
B. They show the balance between finding relevant cases and limiting irrelevant alerts
C. They prove that a model can be transferred to any institution
D. They replace the need for investigation

**3. Which is a material risk of psychometric credit assessment?**

A. The assessment cannot produce any data
B. Every applicant will receive the same result
C. Language, education or familiarity with the interface may influence the score
D. Conventional credit information becomes unlawful

**4. Why should a cluster-based portfolio be tested on later, unused data?**

A. To assess whether the historical grouping remains useful beyond the sample used to create it
B. To ensure that the clusters never change
C. To remove transaction costs and liquidity constraints
D. To turn an unsupervised method into a labelled classification task

**Correct answers:** 1—B; 2—B; 3—C; 4—A.

# Recommended Reading

There is no specific recommended reading for this topic, beyond the case studies.

# Summary

The three cases concern different decisions and use different evidence. Aurora combines a wider view of payment relationships to improve financial-crime detection; psychometric scoring supplements limited conventional credit records; and portfolio clustering groups instruments without an existing target label.

None of these methods is a complete solution on its own. The data boundary, baseline, implementation process and cost of error determine whether the result is useful. A promising experimental result therefore supports further controlled evaluation rather than automatic adoption.

# Self-Assessment Questions

The following questions are not graded. Use them to compare the cases and test whether their findings support the proposed decisions.

1. Why did Project Aurora test national and cross-border views in addition to institution-level monitoring, and what new risks does collaboration introduce?
2. Explain why precision and recall are both relevant when suspicious cases are rare.
3. Identify two ways in which a psychometric credit assessment might disadvantage an applicant for reasons unrelated to creditworthiness.
4. Why is out-of-sample testing especially important when clusters are used to construct a financial portfolio?
5. Select one case and explain what evidence would be required before using its approach in a South African financial institution.

# Glossary of terms

The glossary provides brief definitions of key terms used in Topics 1 to 6.

**Accountable institution:** A business or profession subject to specified duties under the FIC Act.

**Algorithmic trading:** Use of programmed rules to generate, route or execute trades.

**Application programming interface (API):** A defined interface through which software systems exchange data or services.

**Baseline:** A simple or current method used to judge whether a proposed approach performs better.

**Batch processing:** Processing a defined collection of data at a scheduled time or interval.

**Big data:** Data requiring scalable storage and processing because of its volume, variety, velocity or variability.

**Borrower:** A person or organisation that receives funds and must repay them.

**Calibration:** The degree to which predicted probabilities match observed outcome frequencies.

**Computational notebook:** An interactive document combining executable code, results and explanatory text.

**Computer vision:** Methods for extracting or evaluating information from images or video.

**Counterparty:** The other party to a financial contract or transaction.

**Cryptography:** Mathematical methods for protecting information, verifying authorisation or detecting alteration.

**Data lake:** A store for large volumes of varied data, often before detailed structuring.

**Data lineage:** The traceable path from data's origin through transformation to its final use.

**Data profiling:** Systematic examination of a dataset's structure, values, completeness and distributions.

**Data warehouse:** A curated store of structured data for analysis, reporting and reuse.

**Dependency management:** Recording and controlling the external software packages required by analytical code.

**DevOps:** Practices connecting software development and operations to support reliable testing and release.

**Enterprise resource planning (ERP):** Integrated software for managing processes such as accounting, procurement and human resources.

**Entity resolution:** Determining whether records from different sources refer to the same entity.

**Event streaming:** Capturing, processing and routing events continuously as they are produced.

**Extract-load-transform (ELT):** Loading source data before transforming it in the target platform.

**Extract-transform-load (ETL):** Transforming source data before loading it into the target platform.

**F1 score:** The harmonic mean of precision and recall.

**False negative:** A relevant case incorrectly classified as negative.

**False positive:** An irrelevant case incorrectly classified as positive.

**Fintech:** Use of technology to provide or support financial services.

**High availability:** The ability of a system to remain accessible when required.

**Imputation:** Replacing a missing value with an estimated or assigned value.

**Index:** A data structure that helps a system locate selected records efficiently.

**Information leakage:** Use of information during model development that would be unavailable at decision time.

**Latency:** The delay between receiving an input and producing the required output.

**Legacy system:** An older operational system that may be difficult to change or integrate.

**Machine learning:** Methods that learn patterns from data and apply them to new observations.

**Machine-learning operations (MLOps):** Practices for controlled model deployment, monitoring, maintenance and retraining.

**Metadata:** Information describing data's meaning, format, origin, ownership or permitted use.

**Model risk:** Risk arising when a model is incorrect, misused or applied beyond its limitations.

**Natural language processing (NLP):** Methods for analysing, representing or generating human language.

**Objective function:** A mathematical expression that an optimisation procedure seeks to maximise or minimise.

**On-premises infrastructure:** Computing and storage hosted in facilities controlled by the organisation.

**Operational risk:** Risk of loss from failed processes, people, systems or external events.

**Optimisation:** Selecting the best feasible action under a defined objective and constraints.

**Partitioning:** Dividing a dataset into smaller logical parts for storage or processing.

**Precision:** The proportion of flagged cases that are genuinely relevant.

**Predictive analytics:** Using available information to estimate an unknown or future outcome.

**Proof of concept:** A limited test of whether an idea or approach is feasible.

**Psychometric assessment:** A structured instrument for measuring abilities, attitudes or other psychological characteristics.

**Random seed:** A starting value used to reproduce a procedure involving randomness.

**Real-time analytics:** Producing an output quickly enough to influence the event or decision being observed.

**Recall:** The proportion of all relevant cases that a model identifies.

**Schema:** A defined representation of data fields, structures, relationships and rules.

**Segregation of duties:** Separating incompatible responsibilities to provide independent review and control.

**Semi-structured data:** Data with labels or markers but no single fixed tabular structure.

**Structured data:** Data organised according to a predefined structure.

**Structured Query Language (SQL):** A language for defining, querying and manipulating relational data.

**Tokenised asset:** A digital representation of ownership or rights associated with an asset.

**Trading venue:** A regulated or organised facility through which financial instruments are traded.

**Unstructured data:** Data whose meaning is not stored in predefined fields.

**Version control:** A system for recording, comparing and reviewing changes to code or documents.

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# Answers to Self-Assessment Questions

## TOPIC 1 SELF-ASSESSMENT ANSWERS

1. For the purposes of this module, the principal sub-sectors are banking, insurance and asset management, with fintech operating across them. Banking produces payment, credit and account data; insurance produces policy, premium and claims data; and asset management relies heavily on portfolio, company and market data. Other valid industry classifications may include additional areas.

2. It matters because the meaning and use of data differ across sub-sectors. Retail banking may require a fraud score while a card transaction is in progress, whereas an insurer may analyse claims that develop over a longer period. The model, time horizon and evaluation measure should be selected according to the specific business process.

3. Blockchain may support shared transaction records, tokenised assets and smart contracts. These applications create data relating to transactions, ownership and automated events. A data scientist might evaluate settlement patterns, identify unusual activity or test whether an automated trigger produced the intended result. The usefulness of the analysis still depends on the quality and governance of the input data.

4. The FIC Act requires accountable institutions to implement risk-based controls including customer due diligence, record keeping, transaction monitoring and regulatory reporting. FAIS regulates specified advisory and intermediary services. Applicable providers, key individuals and representatives must meet fit-and-proper requirements that may include qualifications, experience, regulatory examinations and training. A model may organise evidence or prioritise cases, but it does not become the authorised adviser or assume professional responsibility. POPIA also affects personal information and certain solely automated decisions. Together, these requirements influence how financial data and model-supported decisions are governed.

5. A legacy system is an older system that remains important to current operations. Problems arise where integration is limited, knowledge is scarce or definitions differ across systems. A data scientist may consequently spend substantial time tracing fields, reconciling identifiers and checking whether records are comparable before modelling begins.

6. An internal system holds data generated by the institution's own processes, such as accounts, policies, transactions or ledgers. An external provider such as Bloomberg or LSEG supplies market and reference data gathered from outside the institution. The two may be combined, subject to licensing, access and data-governance requirements.

## TOPIC 2 SELF-ASSESSMENT ANSWERS

1. Credit risk concerns the possibility that a borrower or counterparty will not meet an obligation and may draw on repayment and affordability data. Market risk concerns losses arising from market movements and may draw on prices, positions and volatility. Operational risk concerns losses resulting from failed processes, people, systems or external events and may draw on incident and control data. For example, a home-loan default model concerns credit risk, whereas a scenario estimating a bond portfolio's loss after an interest-rate change concerns market risk.

2. Segmentation groups customers into groups with similar characteristics (for example, by income stability and spending category) to target each group differently; personalisation goes further, tailoring a product or price to an individual customer's specific data. A bank might use segmentation to design a handful of product bundles, and personalisation to set an individual customer's specific interest rate.

3. Fraud systems have to balance missed fraud against the cost of incorrectly blocking genuine activity. Lowering the threshold usually catches more suspicious cases, but

also creates more customer disruption and investigation work; raising it has the opposite effect. The appropriate balance depends on the severity and reversibility of each error and has to be reflected in both model evaluation and operational review.

4. Sentiment analysis extracts a measure from text such as news, company announcements or social-media posts. This information differs from price and volume data because it describes language and expressed opinion. It may provide a complementary input, but its usefulness depends on the market, period, source and modelling approach.

5. Real-time analytics is described as a technical requirement because it is the capability, processing data and producing a decision within seconds, that other application areas (fraud detection, algorithmic trading, dynamic pricing) depend on to be useful; it is not, on its own, a distinct business use case. For example, a fraud score is only valuable if it is available before a transaction completes, not in a report the next day.

6. Predictive analytics is the broader use of available information to estimate an unknown or future outcome. Forecasting focuses on quantities indexed over time. Fraud detection may classify a transaction taking place now, whereas cash-flow planning forecasts a sequence of future values. Pricing may estimate expected claim cost without producing a time series. The method should therefore be selected according to the business outcome, timing and available data.

## TOPIC 3 SELF-ASSESSMENT ANSWERS

1. The first step is to determine the target outcome, time of decision, decision and unit of analysis. Then each suggested source needs to be verified for ownership, meaning, quality, update frequency, lawful purpose, access rights and retention and whether the field will even exist in production. Collecting all the fields you can get, raises costs and privacy exposure, and may add unrelated or leaked information. A defensible dataset is one that has data that are relevant, permitted and understood.

2. Policy tables are designed because they conform to predefined fields and relationships. Adjuster notes are unstructured free text, but may have structured metadata associated with it. Photos are disorganised images. A controlled design would keep originals, associate each item with a claim thru governed identifiers, record provenance and access, and generate derived text or image features in a reproducible pipeline. Team members should verify that the labels are consistent and run tests to see if the performance varies due to language, image quality, or document type.

3. The notebook only says that an experiment was successful in one environment. Production deployment must also include independent evaluation, controlled and tested code, reproducible dependencies, secure data access, sufficient capacity, integration with the operational process, monitoring, fallback procedures, ownership and change control. The team needs to figure out how the score drives actions, how incorrect alerts impact customers, and when human review is necessary. The model should be validated on the population for which it is intended and monitored for deterioration after release.

4. Precision and recall against complaints reviewed by a person could be one technical measure. Business measures could include time to first response, time to resolution, repeat contact and upheld complaints. The team should also determine whether short messages, or messages that are written in a poorly represented language, are given a lower priority. A useful plan includes the baseline and target, frequency of monitoring, results by relevant group and the route for reviewing adverse cases.

5. Documentation records the purpose, data, assumptions, implementation, results and limitations needed to reproduce and interpret a model. Independent challenge tests credibility of those claims and may reveal leakage. Unstable relationships. Coding errors or inappropriate use. The apparent strong performance cannot be trusted without these controls and subsequent users may use the model outside its tested scope. They improve the technical evidence and quality of decisions, but also offer accountability and continuity.

6. The data scientist should present the evidence, not hide or downplay it. The analysis and affected groups should be documented and raised thru the project's governance route which may include the model and business owners, independent validation, risk, compliance, legal, privacy and executive decision makers. The response may involve redesigning variables or labels, changing thresholds, adding human review, limiting use or stopping deployment. If the normal route does not work, and there is still a likelihood of material harm, the institution's formal ethics, risk or whistleblowing process should be followed.

## TOPIC 4 SELF-ASSESSMENT ANSWERS

1. Big data is characterised not by a single row-count threshold but by the pressure on the architecture. Velocity. Transactions come in continuously here and have to be processed before they are authorised, even if the data set is small. The system also has to deal with duplicates, late events and failures. The volume becomes more important when the same transactions are kept for investigation, reporting and later model development.

2. Streaming can inspect a transaction before authorisation, update short term behavioural features and provide an instant response. Other things like reconciling

settled transactions, refreshing longer term features, retraining models and generating control reports are for batch processing. Keeping both paths allows a fast decision to be followed by thorough and repeatable checks. Their identifiers, event times and definitions must be the same so that the earlier decision can be reproduced later.

3. A lake can offer flexible and scalable storage, but “move everything” is not a governance strategy. The institution needs to identify valuable use cases, permitted purposes and retention periods; catalogue ownership, meaning and lineage; and apply quality controls appropriate for each dataset. Even when direct identifiers are removed, access restrictions must be in place to protect personal data. The cloud provider’s security, resilience, recovery, jurisdiction, exit arrangements and concentration risk must be evaluated. Curated layers are still needed so that users are not unknowingly analysing duplicated, obsolete or incompatible records.

4. A bigger sample can reproduce historical bias, have incorrect labels or it can reflect a process that no longer works. There may be time-dependence in financial observations, so random splitting of records may leak future conditions into training. Markets in flux offer more opportunities to fit noise with additional variables. quality depends on sample being representative of intended population and decision time, labels being representative of outcome required, evaluation being reflective of future use. Learning only happens when these design conditions are sound.

5. The team should first determine if the records are of the same entity and what balance concept and time stamp each system uses. Integration can involve mapping of identifiers, currencies, time zones, product definitions and effective dates. Totals should reconcile to trusted sources and unmatched or conflicting records should be examined. Entity-resolution rules should be tested for false and missed matches. The final dataset should include source fields, transformation rules, quality exceptions and lineage to enable a reported or modelled balance to be traced.

6. The card system generates a transaction event that includes governed identifiers and transaction attributes . The event-streaming service feeds the event to feature and model-scoring components which combine it with authorised recent history and return a decision within the required time. Events and decisions are recorded permanently for investigation and replay. Batch or scalable distributed processing of longer histories, monitoring data, retraining datasets. Curated structured storage for reporting. You need controls for access, encryption, lineage, schema changes, model versioning, latency, duplicate events, fallback behaviour, validation, customer impact monitoring.

## TOPIC 5 SELF-ASSESSMENT ANSWERS

1. An appropriate statement is: “Which new loans should be approved, referred or declined at the moment of application so that twelve-month credit loss decreases without unfairly excluding reliable borrowers?” The unit is one application, the outcome is defined arrears or loss within twelve months, the observation point is the application decision and the action is approve, refer or decline under an approved policy.

2. Some problems need reliable description, reconciliation or process repair, not prediction. If two systems have different cut-off times and this causes branch totals to differ, it is more useful to agree on the definition and adjust the integration process than it is to fit a model.

3. Zero may indicate a real value, a missing response, a system default or “not applicable”. The analyst will look at the data dictionary and capture process, compare related fields, talk to the owner and determine how many records are impacted. Any correction, treatment of missing values or exclusion must be documented and tested.

4. Prediction estimates the likelihood of a transaction being fraudulent. Optimisation identifies which transactions should be blocked or reviewed, taking into account loss, customer inconvenience, investigator capacity and other constraints. A good estimate alone does not tell what is the best action.

5. The institution requires a baseline prior to the intervention, a suitable comparison/controlled rollout, and outcome measures that reflect the decision. These may include loss avoided, false declines, review effort, customer complaints and results by relevant group. Conditions and behaviour may change, so ongoing surveillance is required.

## TOPIC 6 SELF-ASSESSMENT ANSWERS

1. Criminal networks can operate across institutions and borders, so a broader perspective can reveal links that no single institution can see. Collaboration also creates legal, privacy, security, governance and accountability risks and must be enabled by controlled data access and clear authority.

2. Recall is the amount of suspicious activity found; precision is the amount of relevant alerts. Investigators can be swamped by false alerts if recall is maximised, while maximising precision can mean that important activity is not detected. The balance has to consider the investigative capacity and the consequences of a missed case.

3. Language or educational background, disability, familiarity with the assessment, or comfort with a digital interface may impact performance. These factors can affect the score but do not indicate willingness or ability to pay back.

4. Clusters are based on past relationships that can change. Even decisions as to sample, number of clusters and variables may follow patterns of chance. Testing the resulting portfolio on later, unused data gives a more credible indication of whether the grouping generalises.

5. Legal and representative data, a proper baseline, locally validated performance, analysis of impacted groups, operational capacity, security and privacy controls, and documented approval and monitored rollout would be required from a local entity. The specific evidence depends on the decision and its consequences.